Super interesting title on weakly interacting fermionic systems

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Contents

1 Second Quantization

INTROOOOO

1.1 Principles of quantum mechanics

Preliminaries

In this preliminary section we give an overvew of the mathematical objects that appear in the description of quantum systems. The probabilistic nature of a quantum system is realized in the linear structure of a complex vector space that is equipped with an inner product, which produces the probability amplitude for a system prepared in some state to be measured in another. In this setting, linear combinations of state vectors are just probabilistic superpositions. We will give the necessary definitions regarding a particular class of vector spaces, even though many concepts come from more general settings.

Definition 1.1 (Hilbert space)**.** A (complex) *Hilbert space* $\mathcal{H}$ is a vector space equipped with a sesquilinear map $\langle\cdot,\cdot\rangle : \mathcal{H}\times\mathcal{H}\to\mathbb{C}$ such that $\|\phi\| \coloneqq \langle\phi,\phi\rangle^{1/2}$ defines a norm which makes $\mathcal{H}$ a complete metric space.

A note on the choice of notation. Sometimes, using Dirac's "bra-ket" notation can help avoid ambiguities. In this notation, a vector would be denoted by $|\psi\rangle \in \mathcal{H}$, a linear functional as $\langle\psi| \in \mathcal{H}^*$, and the inner product (with respect to a symmetric operator, which we haven't yet introduced) as $\langle\psi, A\phi\rangle = \langle\psi|A|\phi\rangle$. An example of such a scenario where this notation *might* be preferred is that of the occupation number basis, which we make use of in Chapter (3).

Definition 1.2. A Hilbert space is said to be *separable* if it admits a countable dense subset.

The relevance of this definition stems from the fact that, if we can find a countable set $\mathcal{S} = \{u_n : n\in\mathbb{N}\}$ of vectors in the Hilbert space $\mathcal{H}$ such that the space of *finite* linear combinations of elements from $\mathcal{S}$ is dense, we have that any vector in $\mathcal{H}$ can be expressed as a limit of such finite linear combinations. That is, the set $\mathcal{S}$ is *complete* (acts as a *basis*) in the sense that, ignoring problems with linear independence, any vector can be identified with a sequence of components $\{a_n : n\in\mathbb{N}\}$, with the condition that the series $\sum_{n\in\mathbb{N}} a_n u_n$ converges (in the norm topology of $\mathcal{H}$). When speaking of Hilbert spaces we will always assume that they are separable.

Lemma 1.1. *If X is a dense subspace of the Hilbert space $\mathcal{H}$, it is always possible to choose a complete orthonormal system $\{u_n : n\in\mathbb{N}\}$ for $\mathcal{H}$ composed only of elements in X.*

Proof. Since $\overline{X} = \mathcal{H}$, for all $x\in\mathcal{H}$ there exists a sequence $\{x_n\}_{n\in\mathbb{N}} \subseteq X$ such that $x_n \to x$. Let $\{y_n\}_{n\in\mathbb{N}}$ be a complete orthonormal set for $\mathcal{H}$, then for any n there exists a sequence $\{u_m^{(n)}\}_{m\in\mathbb{N}}$ such that $\lim_m u_m^{(n)} = y_n$. Define

$$\mathcal{S} \coloneqq \{u_m^{(n)} : n, m\in\mathbb{N}\},$$

which is countable because it is a countable union of countable sets. The set $\mathcal{S}$ is complete, indeed let $x\in\mathcal{H}$, then there exists a sequence $\{\lambda_m\}_{m\in\mathbb{N}}$ such that $\sum_{k=1}^{\infty}|\lambda_k|^2 < \infty$ and $x = \sum_{k=1}^{\infty}\lambda_k y_k$: we can express y_k as a limit, as discussed above, to get that

$$x = \sum_{k=1}^{\infty}\lambda_k y_k = \sum_{k=1}^{\infty}\lambda_k \lim_{m\to\infty} u_m^{(k)} = \lim_{m\to\infty}\sum_{k=1}^{\infty}\lambda_k u_m^{(k)},$$

which implies that the sequence $\{\sum_{k=1}^{N}\lambda_k u_m^{(k)}\}_{m,N\in\mathbb{N}} \subseteq X$ converges to x (you sure?). Lastly, recall the infinite-dimensional version of the Grahm-Schmidt orthonormalization procedure, then it is clear that to $\mathcal{S}$ we can associate a complete orthonormal system. □

In quantum mechanics, observables are represented by a special type of linear map on the state space, so let us briefly recall the (purely formal) motivation behind this. If A is a physical observable, the result of an experiment is the collapse of the system onto a state associated to a well-defined value of A. A generic state ψ will be a superposition of such vectors, which we may call $\{\psi_i : i\in\mathbb{N}\}$, and let us call $\{\lambda_i : i\in\mathbb{N}\}\subset\mathbb{R}$ the well-defined values of A associated to such vectors. The generic state ψ then encodes the probability amplitudes for system to collapse in each ψ_i, which suggests that the statistical expectation for the measurement of A, when the system is in the state $\psi = \sum_i \lambda_i\psi_i$, would read something like

$$\langle A\rangle_\psi = \sum_{i\in I}\lambda_i\langle\psi_i,\psi_i\rangle = \langle\psi, A\psi\rangle$$

if we could represent the observable by a linear map $A : \mathcal{H}\to\mathcal{H}$. Let us make this precise and give meaning to what we just wrote.

Definition 1.3 (Operator)**.** A *linear operator* (or simply an *operator*) A on $\mathcal{H}$ is a linear map $A : D(A) \to \mathcal{H}$ defined on a subspace $D(A) \subseteq \mathcal{H}$. If $D(A)$ is dense in $\mathcal{H}$, A is said to be *densely defined.*

We will often write "operator" to mean a densely defined operator on a Hilbert space, unless stated otherwise. Moreover, note that the domain on which an operator acts is part of the definition, and will therefore need to be specified case by case.

Definition 1.4 (Extension of an operator)**.** If A and B are two operators such that $D(A) \subseteq D(B)$ and $A\phi = B\phi$ for all $\phi \in D(A)$, then we say that B *extends* (or is an extension of) A, and we write $A \subseteq B$.

Definition 1.5 (Symmetric operators and observables)**.** An operator A is said to be *symmetric* if $\langle \psi, A\phi \rangle = \langle A\psi, \phi \rangle$ for all $\phi, \psi \in D(A)$.
An *observable* is a physical quantity represented by a symmetric operator on the Hilbert space of physical states.
Given A a symmetric operator and $\psi \in D(A)$, the *expectation of A on ψ* is the quantity

$$\langle A \rangle_\psi \coloneqq \langle \psi, A\psi \rangle.$$

Remark 1. The symmetry condition $\langle \psi, A\phi \rangle = \langle A\psi, \phi \rangle$ for all $\phi, \psi \in \mathcal{H}$ is equivalent to the condition $\langle \psi, A\psi \rangle \in \mathbb{R}$ for all $\psi \in \mathcal{H}$, which in turn is equivalent to asking that expectation values of observables must be real numbers.

Remark 2. We can introduce a partial ordering on symmetric operators. If A and B are symmetric operators defined on a common domain $D \subseteq \mathcal{H}$, then we say that $A < B$ if $\langle \phi, A\phi \rangle < \langle \phi, B\phi \rangle$ for all $\phi \in D$.

Definition 1.6 (Bounded operator)**.** A densely defined operator A is said to be *bounded* if it is continuous with respect to the norm topology, and *unbounded* if it is not bounded.

Remark 3. By linearity, continuity is equivalent to the condition that

$$\|A\| \coloneqq \sup_{\substack{\psi \in D(A) \\ \|\psi\|=1}} \|A\psi\| < \infty,$$

and the quantity $\|A\|$ is called the *operator norm* of A.

Proposition 1.2. *Given $A : D(A) \to \mathcal{H}$ a densely defined bounded operator on $\mathcal{H}$, there exists a unique extension to a bounded operator on the entire space $\tilde{A} : \mathcal{H} \to \mathcal{H}$.*

Proof. Let $\{x_n\}_{n\in\mathbb{N}} \subseteq D(A)$ be a Cauchy sequence in $D(A)$, then there exists a unique $x \in \mathcal{H}$ such that $x_n \to x$. Because A is bounded it holds that $\|Ax_n - Ax_m\| \leq \|A\|\|x_n - x_m\|$ and therefore the sequence $\{Ax_n\}_{n\in\mathbb{N}}$ is also a Cauchy sequence. As such, there exists a unique $y \in \mathcal{H}$ such that $Ax_n \to y$. Define the operator $\tilde{A}$ as

$$\begin{aligned} \tilde{A} : \mathcal{H} &\longrightarrow \mathcal{H} \\ x &\longmapsto \tilde{A}x \coloneqq \lim_n Ax_n \end{aligned}$$

where $\{x_n\}_{n\in\mathbb{N}}$ is a Cauchy sequence that converges to x. $\tilde{A}$ extends A and is defined on the entire space. Moreover, it is bounded as $\|A\| < \infty$ and

$$\|\tilde{A}x\| = \|\lim_n Ax_n\| = \lim_n \|Ax_n\| \leq \lim_n \|A\|\|x_n\| = \|A\|\|x\|.$$

□

It is worth underlying that from Proposition (1.2) follows that for bounded operators it is possible to deal with dense subspaces without losing generality. In particular, since we are always assuming the Hilbert space to be separable, it will suffice to define the image of only the basis elements.

Definition 1.7 (Adjoint)**.** If A is an operator on $\mathcal{H}$ we define the *adjoint* A^* of A as the operator $A^* : D(A^*) \to \mathcal{H}$ defined on

$$D(A^*) = \{\phi \in \mathcal{H} : \sup_{\substack{\psi \in D(A) \\ \|\psi\|=1}} |\langle \phi, A\psi \rangle| < \infty\}$$

and with $A^*\phi$ defined for all $\phi \in D(A^*)$ as

$$\langle A^*\phi, \psi \rangle = \langle \phi, A\psi \rangle \quad \forall \psi \in D(A).$$

If $A^* = A$ then A is said to be *self-adjoint*, and *Hermitian* if both operators are defined on the entire Hilbert space.

Proposition 1.3. *The adjoint of a bounded operator is a bounded operator.*

Proof. CHECK. We are only considering densely defined operators, therefore, without loss of generality, let $A : \mathcal{H} \to \mathcal{H}$ be a bounded operator on the whole space $\mathcal{H}$. For all $\phi \in D(A^*)$, the map $\psi \mapsto \langle A^*\phi, \psi \rangle$ is a linear functional on $\mathcal{H}$, where

$$D(A^*) = \left\{ \phi \in \mathcal{H} : \sup_{\|\psi\|=1} \langle \phi, A\psi \rangle < \infty \right\}.$$

Then, by Schwarz's inequality, we have that for every $\phi \in D(A^*)$

$$\|A^*\phi\| = \sup_{\|\psi\|=1} |\langle A^*\phi, \psi \rangle| = \sup_{\|\psi\|=1} |\langle \phi, A\psi \rangle| \leq \sup_{\|\psi\|=1} \|A\psi\| \|\phi\|,$$

where the first step makes use of Riesz representation theorem, which states that the norm of the vector $A^*\phi$ is equal to the norm of the functional we previously wrote. It follows from the boundedness of A that

$$\|A^*\phi\| \leq \|\phi\| \|A\| < \infty$$

which implies that A^* is bounded. □

Proposition 1.4. *An operator A is symmetric if, and only if, $A \subseteq A^*$.*

We point to Appendix A.1 for a discussion on the most important definitions and properties of special types of operators, and end the introduction of the basic objects used here. A key role in the discussion of quantum systems is played by the space of square integrable functions on $\mathbb{R}^n$

$$L^2(\mathbb{R}^n) = \left\{ f : \mathbb{R}^n \to \mathbb{C} \,\middle|\, \left(\int_{\mathbb{R}^n} \mathrm{d}x \, |f(x)|^2 \right)^{1/2} < \infty \right\}$$
$$\text{with} \quad \langle f, g \rangle \coloneqq \int_{\mathbb{R}^n} \mathrm{d}x \overline{f(x)} g(x),$$

whose elements are the wavefunctions of a physical system. Its relevance stands from the fact, that in this space, the squared norm of a state $\psi \in L^2(\mathbb{R}^n)$ makes it possible to write probability amplitudes as integrals of a *probability density*, given as the point-wise modulus squared of the wave function: $|\psi(x)|^2$.

Going back to the definition of expectation value of an observable, we can now make an important generalization. Suppose the system has probability p_i of being prepared in some initial state ψ_i, of course given the normalization condition $\sum_i p_i = 1$, then the expectation of an observable A would be the weighted average of the expectations on each initial state.

Definition 1.8 (Pure and mixed states)**.** A *pure state* is a vector in $\mathcal{H}$, while a *mixed state* is a statistical mixture of pure states given by the collection of probability-state pairs $\{(p_i, \psi_i) : i \in \mathbb{N}\}$, with $\sum_i p_i = 1$. If A is an observable, the expectation of A on a mixed state is defined as

$$\langle A \rangle = \sum_{i \in \mathbb{N}} p_i \langle \psi_i, A\psi_i \rangle.$$

We can take this definition further by following another formal argument. Assuming the observable A is a compact self-adjoint operator, by the spectral theorem we can write it as a series of projectors onto its eigenspaces. Let $\{\phi_\lambda : \lambda \in \sigma(A)\}$ be the eigenvectors of A, then

$$A = \sum_{\lambda \in \sigma(A)} \lambda \langle \phi_\lambda, \cdot \rangle \phi_\lambda.$$

Using this, we write the expectation of A on the mixed state $\{(p_i, \psi_i)\}$ as

$$\langle A \rangle = \sum_{i \in I} p_i \langle \psi_i, A\psi_i \rangle = \sum_{i \in \mathbb{N}, \lambda \in \sigma(A)} p_i \langle \psi_i, \lambda \langle \phi_\lambda, \psi_i \rangle \phi_\lambda \rangle = \sum_{\lambda \in \sigma(A)} \lambda \langle \phi_\lambda, \rho \phi_\lambda \rangle = \mathrm{Tr}(\rho A)$$

where in the last step we defined the *density matrix* which uniquely identifies the mixed state

$$\rho \coloneqq \sum_i p_i \langle \psi_i, \cdot \rangle \psi_i.$$

This argument brings us to a general definition of a quantum state. Recall that $\mathfrak{B}(\mathcal{H})$ is the C^*-algebra of bounded operators on $\mathcal{H}$ (see Appendix A.1).

Definition 1.9. A general quantum state is an operator ρ on $\mathcal{H}$ such that

- $\mathrm{Tr}(\rho) = 1$ (probability normalization condition)
- $\rho^* = \rho$ (self-adjointness)
- $\rho \geq 0$ (positivity of probabilities)

and it defines the *expectation functional*

$$\begin{aligned} \langle \cdot \rangle_\rho : \mathfrak{B}(\mathcal{H}) &\longrightarrow \mathbb{C} \\ A &\longmapsto \langle A \rangle_\rho \coloneqq \mathrm{Tr}(\rho A). \end{aligned}$$

There is a bijection between the set of mixed states (pure states are just a special case) and the set of quantum states (i.e. a subset of the dual of $\mathfrak{B}(\mathcal{H})$). We are interested in the state the system is found in when at thermal equilibrium, and from statistical mechanics we know this state is such that the energy spectrum is populated according to Boltzmann's distribution $p(\epsilon) = e^{-\beta\epsilon}/Z$. With the aforementioned bijection we may identify such a state *at a given temperature* with the mixture $\{(e^{-\beta\epsilon_j}/Z, \psi_j)\}$ if ψ_j are energy eigenstates. To make this precise we need to properly define the foundamental thermodynamic quantities required, which also brings us to the main focus of our work: the free energy, which we will define starting from the known thermodynamic relation $F = U - TS$.

Definition 1.10 (Von Neumann entropy)**.** In the same notation used so far, the *Von Neumann entropy* of a state $\langle \cdot \rangle_\rho$ is defined as

$$S(\langle \cdot \rangle_\rho) \coloneqq -\sum_{n=1}^{\infty} p_n \log(p_n),$$

where we are using the convention that $t \log(t) = 0$ if $t = 0$.

Definition 1.11 (Stability and ground state)**.** A physical system governed by a Hamiltonian, i.e. energy operator, H on $\mathcal{H}$ is said to be *stable* if

$$\inf_{\phi \in D(H), \|\phi\|=1} \langle \phi, H\phi \rangle > -\infty.$$

We define the *ground state energy* as

$$E = \inf_{\phi \in D(H), \|\phi\|=1} \langle \phi, H\phi \rangle,$$

and the *ground state*, **if it exists**, as a (normalized) minimizer for the energy expectation, i.e. a unit vector $\psi_0 \in D(H)$ such that $\langle \psi_0, H\psi_0 \rangle = E$.

Remark 4. If the ground state ψ_0 exists and the ground state energy is E then it holds that $H\psi_0 = E\psi_0$, i.e. ψ_0 is an eigenvector of the energy operator with eigenvalue equal to the ground state energy.

Definition 1.12 (Free energy)**.** The *free energy* of a stable system at temperature $T \geq 0$ is defined as

$$F(T) \coloneqq \inf_{\substack{\langle \cdot \rangle_\rho \\ \langle H \rangle_\rho < \infty}} (\langle H \rangle - T\, S(\langle \cdot \rangle)),$$

which is possibly $-\infty$. If, for $T > 0$, there exists a minimizer for the free energy, then the minimizing state is called a *Gibbs state at (inverse) temperature* β.

Gibbs states are thermal equilibrium states, and for $\beta < \infty$ they are generally mixed, while at $\beta = \infty$ we find the, usually pure, ground state. We will denote Gibbs states by $\langle \cdot \rangle_\beta$, despite the slight ambiguity between this notation and the one used for a generic mixed state, and we will also specify the Hamiltonian, when needed, by writing $\langle \cdot \rangle_{\beta,H}$.

Remark 5. A system composed of a single free particle with state space $L^2(\mathbb{R})$ and governed by the Hamiltonian $H = -\nabla^2$ is stable but it does not admit a ground state, and $F(T) = -\infty$ for all $T > 0$. However, this Hamiltonian does admit a ground state when restricting the state space to wavefunctions on a compact interval. The stability of free particles is essential, as will become clear in the third chapter.

Proposition 1.5. *If $\langle \cdot \rangle_\beta$ is a Gibbs state at $\beta < \infty$ (i.e. at $T > 0$), then for any bounded self-adjoint operator A we have that*

$$\langle A \rangle = \frac{\mathrm{Tr}(Ae^{-\beta H})}{\mathrm{Tr}(e^{-\beta H})}.$$

In particular, $e^{-\beta H}$ is a trace class operator.

Proof. IMPORTANT (and maybe add discussion of $e^{\pm\tau H_0}$ and of $Ve^{-\tau H_0}$ where V is the potential). □

We refer to the appendix for the definition of trace class operators and point out that such operators are compact, which makes $e^{-\beta H}$ a very well-behaved operator in computations.

1.2 Canonical and grand canonical pictures

So far we talked about a generic (separable) Hilber space, but for the most part we will deal with Hilbert spaces constructed in a specific way: tensor products of Hilbert spaces, which are the natural spaces in which to set physical systems composed of multiple degrees of freedom. We are particularly interested in two special subspaces of a tensor product that encode the *statistics* of the particles considered, a property that describes indistinguishabilty between particles in a quantum system. Once again we point to Appendix (A.3) for a discussion on the construction of tensor products (and direct sums) of Hilbert spaces.

Consider a system of N particles whose physical states belong to the spaces $\mathfrak{h}_1, \ldots, \mathfrak{h}_N$. The possible states of this system are described by the tensor product

$$\mathcal{H}_N = \mathfrak{h}_1 \otimes \cdots \otimes \mathfrak{h}_N.$$

Given the energy operators for each particle $h_1, \ldots, h_N$, we lift these operators to the space $\mathcal{H}_N$ with the identification

$$h_i \longmapsto \mathbb{1} \otimes \cdots \otimes h_i \otimes \cdots \otimes \mathbb{1}.$$

The total energy of the system, given only the "internal" degrees of freedom for each particle and disregarding any interaction, is naturally given by

$$H_N^{\text{in}} = h_1 + \cdots + h_N \tag{1}$$

with

$$D(H_N^{\text{in}}) = \operatorname{span}\{\phi_1 \otimes \cdots \otimes \phi_N : \phi_i \in D(h_i),\ i = 1, \ldots, N\}$$

which is dense in $\mathcal{H}_N$ if $D(h_i)$ is dense in $\mathfrak{h}_i$ for all i.

Theorem 1.6 (ground state of non-interacting (distinguishable) particles)**.** *If the ground state energies of the Hamiltonians $h_1, \ldots, h_N$ are*

$$e_j = \inf_{\phi \in D(h_j), \|\phi\|=1} \langle \phi, h_j \phi \rangle, \quad j = 1, \ldots, N$$

then the ground state energy of H_N^{in} is $\sum_{j=1}^N e_j$. Moreover, if $\phi_1, \ldots, \phi_N$ are the ground states of these same Hamiltonians, then the ground state of H_N^{in} is $\phi_1 \otimes \cdots \otimes \phi_N$.

Proof. If $\Psi \in D(H_N^{\text{in}})$ is a unit vector, then we may write $\Psi = \psi_1 \otimes \Psi_1 + \cdots + \psi_K \otimes \Psi_k$, where $\psi_1, \ldots, \psi_K \in D(h_1)$ and $\Psi_1, \ldots, \Psi_K \in \mathfrak{h}_2 \otimes \cdots \otimes \mathfrak{h}_N$ are orthonormal. Since Ψ is normalized this implies that $\|\psi_1\|^2 + \cdots + \|\psi_K\|^2 = 1$, thus

$$\langle \Psi, h_1 \Psi \rangle = \sum_{i=1}^K \langle \psi_i, h_1 \psi_i \rangle \geq \sum_{i=1}^K \|\psi_i\|^2 e_1 = e_1.$$

This implies that $\langle \Psi, H_N^{\text{in}} \Psi \rangle \geq \sum_{j=1}^N e_j$. Conversely, for every $\varepsilon > 0$ we can choose unit vectors $\phi_j \in D(h_j)$ such that $\langle \phi_j, h_j \phi_j \rangle \leq e_j + \varepsilon$ with $j = 1, \ldots, N$, which implies that, for $\Psi = \phi_1 \otimes \cdots \phi_N$ (still a unit vector),

$$\langle \Psi, H_N^{\text{in}} \Psi \rangle = \sum_{j=1}^N \langle \phi_j, h_j \phi_j \rangle \leq \sum_{j=1}^N e_j + N\varepsilon,$$

which holds if, and only if, $\langle \Psi, H_N^{\text{in}} \Psi \rangle \leq \sum_{j=1}^N e_j$, thus we proved the equality. Furthermore, it is clear that if $\phi_1, \ldots, \phi_N$ are the respective ground state eigenvectors of the N one-body Hamiltonians, their tensor product is an eigenvector of H_N^{in} with eigenvalue given precisely by $\sum_{j=1}^N e_j$, the ground state energy. □

If we consider interactions we need to lift additional operators. For pair-wise interactions (but this procedure can be easily generalized to N-body interactions) to each pair (i, j) of distinct particles $(i \neq j)$ we associate an operator W_{ij} on a suitable domain contained in $\mathfrak{h}_i \otimes \mathfrak{h}_j$. To lift this operator to $\mathcal{H}_N$ we again make use of the identification

$$W_{ij} \longmapsto \mathbb{1} \otimes \cdots \otimes W_{ij} \otimes \cdots \otimes \mathbb{1},$$

which gives a tensor product of $N-2$ identity operators and one interaction operator. Without specifying the correct domain we arrive at the formal way of writing the Hamiltonian of a system of N pair-wise interacting particles:

$$H_N = H_N^{\text{in}} + \sum_{1 \leq i < j \leq N} W_{ij} = \sum_{j=1}^{N} h_j + \sum_{1 \leq i < j \leq N} W_{ij}. \tag{2}$$

Requiring that particles be indistinguishable from one another means that both their internal energy operators and the pair-wise interactions do not depend on the specific particle considered. This means that

$$\begin{aligned} &\mathfrak{h}_1 = \ldots = \mathfrak{h}_N =: \mathfrak{h}, \\ &h_1 = \ldots = h_N =: h, \\ &W_{ij} = W_{kl} =: W \quad \forall i \neq j, k \neq l. \end{aligned}$$

When lifting these operators to $\mathcal{H}_N$ we still need to specify indices that identify the specific factor of the tensor product they act on, for example

$$h_1 = h \otimes \mathbb{1} \otimes \cdots \otimes \mathbb{1}, \quad W_{12} = W \otimes \mathbb{1} \otimes \cdots \otimes \mathbb{1}.$$

Indistinguishability also requires an additional property that physical states must satisfy, so let us take the time to recall a formal explanation of what this property is. Consider the simplest case of a system of two indistinguishable particles in the states $\psi, \phi \in \mathfrak{h}$. Because quantum states can be told apart only through probability amplitudes, two vectors that differ only by an overall phase can be considered indistinguishable, as $|\langle \phi, e^{i\theta}\phi \rangle|^2 = \|\phi\|^2$ no matter the phase θ. Therefore we can consider an exchange operator that flips the order in a simple tensor in $\mathfrak{h} \otimes \mathfrak{h}$

$$\text{Ex}(\phi \otimes \psi) = \psi \otimes \phi$$

and say that two indistinguishable particles only admit physical states that differ at most by an overall phase from their exchanged version, i.e. there must exist some α such that

$$\text{Ex}(\phi \otimes \psi) = e^{i\alpha} \phi \otimes \psi.$$

But since Ex^2 should act as the identity, as flipping two times the order of the particle states results in the original state, we can conclude that

$$\text{Ex}^2 = \mathbb{1} \implies e^{2i\alpha} = 1 \implies \alpha = \pm 1.$$

The sign $\alpha = \pm 1$ is what determines the statistics of the two identical particles: *bosonic* (+) and *fermionic* (-). Although we will only focus on explicit calculations in the fermionic case, the statistical differences between bosons (particles that follow bosonic statistics) and fermions (particles that follow fermionic statistics) can be huge and have extremely deep physical consequences.

We will now make precise the notion of bosonic and fermionic physical states. On the space $\mathcal{H}_N$ we can define a natural action for the group of permutations S_N. Indeed, we can build the unitary representation U_σ of $\sigma \in S_N$ on this space by specifying the action on the usual basis (see Appendix A.3):

$$U_\sigma(u_1 \otimes \cdots \otimes u_N) \coloneqq u_{\sigma^{-1}(1)} \otimes \cdots \otimes u_{\sigma^{-1}(N)}.$$

Define the operators

$$\begin{aligned} P_+ &\coloneqq \frac{1}{N!} \sum_{\sigma \in S_N} U_\sigma \\ P_- &\coloneqq \frac{1}{N!} \sum_{\sigma \in S_N} \operatorname{sgn}(\sigma) U_\sigma. \end{aligned} \tag{3}$$

These are unitary by construction and one can show that they are also projections.

Definition 1.13 (Symmetric and anti-symmetric tensor products)**.** Given $\mathfrak{h}$ the one-particle Hilbert space, the *symmetric tensor product* is the subspace

$$\bigotimes_{sym}^{N} \mathfrak{h} \coloneqq P_+ \left(\bigotimes^{N} \mathfrak{h} \right) \subset \mathcal{H}_N$$

while the *anti-symmetric tensor product* is the subspace

$$\bigwedge^{N} \mathfrak{h} \coloneqq P_- \left(\bigotimes^{N} \mathfrak{h} \right) \subset \mathcal{H}_N.$$

We also define the anti-symmetric tensor product of vectors as

$$u_1 \wedge \cdots \wedge u_N = \sqrt{N!}\, P_-(u_1 \otimes \cdots \otimes u_N).$$

Bosonic physical states belong in the symmetric tensor product, while fermionic physical states belong in the anti-symmetric tensor product.

Remark 6. In the special case where $\mathfrak{h} = L^2(\mathrm{R}^n)$, anti-symmetric tensor products between wave-functions can be constructed as Slater determinants, i.e.

$$u_1 \wedge \cdots \wedge u_N(x_1, \ldots, x_N) = \frac{1}{\sqrt{N!}} \det \begin{pmatrix} u_1(x_1) & \cdots & u_N(x_1) \\ \vdots & & \vdots \\ u_1(x_N) & \cdots & u_N(x_N) \end{pmatrix}$$

Remark 7. If $\dim(\mathfrak{h}) < N$ then $\bigwedge^N \mathfrak{h} = \{0\}$, i.e. the anti-symmetrization of N vectors in a space of dimension less than N is always the null vector. This is especially important because physical states of fermionic systems live in the anti-symmetric tensor product, and if the Hilbert space of a single particle is finite-dimensional the allowed number of particles will be finite.

With the restrictions we put on the N-body Hamiltonian, it is clear that H_N maps the spaces $\bigotimes_{\text{sym}}^N \mathfrak{h}$ and $\bigwedge^N \mathfrak{h}$ into themselves, which means that we can restrict this N-body Hamiltonian to either of the two subspaces and get a well-defined operator (provided that a suitable domain for the starting Hamiltonian is chosen). The operators $P_+(H_N)$ and $P_-(H_N)$ are respectively called the *bosonic* and *fermionic* N-body Hamiltonians for interacting indistinguishable particles.
This way of defining the energy of the system characterises the *canonical picture*, where the number of particles is fixed. But in statistical mechanics, allowing the number of particles to vary can actually make things easier. Allowing N to vary leads to the definition of the *grand canonical picture*. We shall now discuss the state space for a system in the grand canonical picture, which we can define quite easily recalling the results mentioned in Appendix A.3.

Definition 1.14 (Fock space)**.** For every $N \in \mathbb{N}$, let $\mathfrak{h}_1, \dots, \mathfrak{h}_N$ be one-particle Hilbert spaces, the *Fock space* is defined as

$$\mathcal{F} := \bigoplus_{N=0}^{\infty} \bigotimes_{i=1}^{N} \mathfrak{h}_i$$

where, as a convention, we interpret $\mathfrak{h}_1 \otimes \cdots \otimes \mathfrak{h}_N$ for $N = 0$ to be just $\mathbb{C}$. The state $1 \in \mathbb{C}$ is called the *vacuum state* and often denoted $|0\rangle$, $|\Omega\rangle$ or Ω.

On this space, the energy operator is naturally given by the sum of all the possible N-body Hamiltonians:

$$H = \bigoplus_{N=0}^{\infty} H_N, \quad \text{with } H \bigoplus_{N=0}^{\infty} \Psi_N = \bigoplus_{N=0}^{\infty} H_N \Psi_N,$$

with H_0 taken to be the null operator (which means we are viewing the vacuum as having zero energy), and its domain is

$$D(H) = \left\{ \Psi = \bigoplus_{N=0}^{\infty} \Psi_N \in \mathcal{F} : \Psi_N \in D(H_N), \ \sum_{N=0}^{\infty} \|H_N \Psi_N\|^2 < \infty \right\}.$$

Introducing indistinguishability in this picture means repeating the steps we took for the N-body case in each *N-particle sector*, which gets us to the definition of the *fermionic* and *bosonic* Fock spaces. Given $\mathfrak{h}$ the state space for a single particle, these spaces are, respectively

$$\mathcal{F}^{\mathrm{F}}(\mathfrak{h}) := \bigoplus_{N=0}^{\infty} \bigwedge^{N} \mathfrak{h}$$
$$\mathcal{F}^{\mathrm{B}}(\mathfrak{h}) := \bigoplus_{N=0}^{\infty} \bigotimes_{\mathrm{sym}}^{N} \mathfrak{h}.$$

One of the most powerful aspects of the grand canonical picture is that all observables of the system can be written in the same way using special operators called *creation/annihilation operators*. Starting from the Fock space $\mathcal{F} = \bigoplus_{N=0}^{\infty} \bigotimes^{N} \mathfrak{h}$ we define operators that add or remove a single particle state $f \in \mathfrak{h}$ as

$$\begin{aligned} a(f)(f_1 \otimes \cdots \otimes f_N) &:= \sqrt{N} \langle f, f_1 \rangle_{\mathfrak{h}} f_2 \otimes \cdots \otimes f_N, \\ a^*(f)(f_1 \otimes \cdots \otimes f_N) &:= \sqrt{N+1} f \otimes f_1 \otimes \cdots \otimes f_N. \end{aligned} \tag{4}$$

Here, for every $f \in \mathfrak{h}$, $a(f)$ is called *annihilation operator* and $a^*(f)$ is called *creation operator.* Note the very important special case where an annihilation operator acts on the vacuum: $a(f)|0\rangle = 0$ for all $f \in \mathfrak{h}$, i.e. annihilation operators annihilate the vacuum. To indicate an operator that could be either of the two, we write $a^{\#}(f)$. With this definition the domain would be the set of pure tensors, but by linearity we can extend this to the space $\cup_{M=0}^{\infty} \bigoplus_{N=0}^{M} \mathcal{H}_N$, which is dense in $\mathcal{F}$.

Proposition 1.7. *Creation and annihilation are the adjoint of one another, i.e. for all $f \in \mathfrak{h}$ and for all $\Psi, \Phi \in \cup_{M=0}^{\infty} \bigoplus_{N=0}^{M} \mathcal{H}_N$ it holds that*

$$\langle a(f)\Psi, \Phi \rangle_{\mathcal{F}} = \langle \Psi, a^*(f)\Phi \rangle_{\mathcal{F}}.$$

Proof. It suffices to prove the equality for simple tensors, so let $\psi_1 \otimes \ldots \otimes \psi_N \in \bigotimes^N \mathfrak{h}$ and $\phi_0 \otimes \phi_1 \otimes \ldots \otimes \phi_N \in \bigotimes^N \mathfrak{h}$. Then, for all $f \in \mathfrak{h}$,

$$\begin{aligned} \langle \psi_1 \otimes \cdots \psi_N, a(f)\phi_0 \otimes \cdots \otimes \phi_N \rangle &= \sqrt{N+1} \langle f, \phi_0 \rangle \langle \psi_1 \otimes \cdots \otimes \psi_N, \phi_1 \otimes \cdots \otimes \phi_N \rangle \\ &= \sqrt{N+1} \langle f, \phi_0 \rangle \prod_{j=1}^{N} \langle \psi_j, \phi_j \rangle, \end{aligned}$$

and similarly

$$\langle a^*(f)\psi_1 \otimes \cdots \otimes \psi_N, \phi_0 \otimes \cdots \otimes \phi_N\rangle = \sqrt{N+1}\langle f, \phi_0\rangle \prod_{j=1}^{N} \langle \psi_j, \phi_j\rangle.$$

□

Once again we introduce indistinguishability into the picture. Looking at the definition of these operators, it is clear that if a state $\Psi_N \in D(a^\#(f))$ is completely symmetric or completely anti-symmetric then applying an annihilation operator produces a state that is still completely symmetric or anti-symmetric (as Ψ_N can be written as a linear combination of all $N!$ permutations of N states and thus $a(f)$ removes all of these states exactly once), but applying a creation operator does introduces a state that is not yet accounted for in the (anti-)symmetrization. We then define *bosonic (+)* and *fermionic (-)* creation/annihilation operators as the projection onto the correct subspaces:

$$a_\pm(f) \coloneqq a(f),\ a^*_\pm(f) \coloneqq P_\pm(a^*(f)) \quad \forall f \in \mathfrak{h}.$$

Remark 8. Bosonic and fermionic creation/annihilation operators are still densely defined and still formally adjoint to one another (within their respective statistics).

Remark 9. The map $\mathfrak{h} \ni f \mapsto a^*_\pm(f)$ is linear but $\mathfrak{h} \ni f \mapsto a_\pm(f)$ is conjugate linear.

Remark 10. The normalization we used in (4) changes slightly when projecting the image onto the (anti-)symmetric subspaces. Indeed it can be shown that

$$a_\pm(f)P_\pm(f_1 \otimes \cdots \otimes f_N) = \sqrt{N+1} \sum_{\sigma \in S_N} \langle f, f_{\sigma(1)}\rangle f_{\sigma(2)} \otimes \cdots \otimes f_{\sigma(N)},$$
$$a^*_\pm(f)f_1 \otimes \cdots \otimes f_N = \frac{1}{\sqrt{N}} \sum_{\sigma \in S_N} \sum_{j=1}^{N+1} f_{\sigma(1)} \otimes \cdots \otimes \underset{(j-\text{th})}{f} \otimes \cdots \otimes f_{\sigma(N)}.$$

Recall the definition of the commutator and anticommutator between two operators defined on a common domain: $\{A, B\} \coloneqq AB + BA$ and $[A, B] \coloneqq AB - BA$.

Proposition 1.8 (CCR/CAR)**.** *The operators a_- and a^*_- satisfy the Canonical Commutation Relations (CCR), while the operators a_+ and a^*_+ satisfy the Canonical Anticommutation Relations (CAR), i.e. for all $f, g \in \mathfrak{h}$ it holds that*

$$\{a_+(f), a^*_+(g)\} = \langle f, g\rangle_{\mathfrak{h}}, \quad \{a_+(f), a_+(g)\} = \{a^*_+(f), a^*_+(g)\} = 0$$
$$[a_-(f), a^*_-(g)\} = \langle f, g\rangle_{\mathfrak{h}}, \quad [a_-(f), a_-(g)] = [a^*_-(f), a^*_-(g)] = 0$$

Proof. EXPLICIT CALCULATION? □

Remark 11. A crucial point, that greatly simplifies the treatment of our system in terms of how well-behaved the operators we are going to use are, is that fermionic creation/annihilation operators are bounded. Let $f \in \mathfrak{h}$, then it is clear that for all normalized $\Psi \in D(a(f))$ it holds that

$$\begin{aligned}\|a_-(f)\Psi\|^2 &= \langle \Psi, a^*_-(f)a_-(f)\Psi\rangle = -\langle \Psi, a_-(f)a^*_-(f)\Psi\rangle + \langle \Psi, \{a_-(f), a^*_-(f)\}\Psi\rangle \\ &= -\langle a^*_-(f)\Psi, a^*_-(f)\Psi\rangle + 1 \leq 1,\end{aligned}$$

which implies that $\|a_-(f)\| \leq 1$, i.e. $a_-(f)$ is bounded for all $f \in \mathfrak{h}$. Similarly for $a^*_-(f)$.

An important quantity we will come back to shortly is the number density operator. Given $u \in \mathfrak{h}$ a unit vector, $\bigotimes^N u$ is an eigenvector with eigenvalue N for the operator $a^*(u)a(u)$, which means this operator "counts" how many times a single-particle state is present in a state in $\mathcal{F}$. This is useful when considering the occupation number basis for a bosonic or fermionic Fock space, with the important distinction that bosonic modes can have arbitrary occupation number, while fermionic modes can only have occupation 0 or 1. Pauli's exclusion principle is realized in quantum mechanics exactly by the antisymmetrization of fermionic states, as, given two one-particle states $\phi, \psi \in \mathcal{H} \setminus \{0\}$, their antisymmetrization $\phi \otimes \psi - \psi \otimes \phi$ is not null if, and only if, $\phi \neq \psi$.

Proposition 1.9. *Let $u_1, \dots, u_r \in \mathfrak{h}$ be unit vectors and $\mathcal{H}_{n_1,\dots,n_r}$ the joint eigenspaces of the operators $a^*_\pm(u_1)a_\pm(u_1), \dots, a^*_\pm(u_r)a_\pm(u_r)$, then the fermionic and bosonic Fock spaces can be decomposed as direct sums of their joint eigenspaces. Indeed:*

$$\mathcal{F}^{\mathrm{B}}(\mathfrak{h}) = \bigoplus_{n_1=0}^{\infty} \cdots \bigoplus_{n_r=0}^{\infty} \mathcal{H}_{n_1,\dots,n_r}$$
$$\mathcal{F}^{\mathrm{F}}(\mathfrak{h}) = \bigoplus_{n_1=0}^{1} \cdots \bigoplus_{n_r=0}^{1} \mathcal{H}_{n_1,\dots,n_r}.$$

Proof. We'll prove the fermionic case, with $\dim(\mathfrak{h}) = M < \infty$ and $r = M$, as this provides the useful fermionic occupation number basis, which will be of great help to us later. Given $\{\phi_1, \dots, \phi_M\}$ an orthonormal basis for $\mathfrak{h}$, and calling $N_j = a^*_-(\phi_j)a_-(\phi_j) = a^*_j a_j$, we can prove that these number operators are commuting projections. These operators are indeed self-adjoint, as

$$N_j^* = (a_j^* a_j)^* = a_j^* a_j = N_j,$$

and idempotent, as the CAR imply

$$N_j^2 = (a_j^* a_j)^2 = -a_j^* a_j^* a_j a_j + a_j^* a_j = a_j^* a_j.$$

Moreover, for $j \neq k$, we have that

$$[N_j, N_k] = a_j^* a_j a_k^* a_k - a_k^* a_k a_j^* a_j = a_k^* a_k a_j^* a_j - a_k^* a_k a_j^* a_j = 0,$$

therefore, all the number operators are commuting projections, furthermore their spectra are all equal to $\{0, 1\}$. Note that these algebraic manipulations are well defined, since all the operators $a^\#$ (and their products) are defined on the entire space $\mathcal{F}^{\mathrm{F}}(\mathfrak{h})$, which is finite-dimensional if $\mathfrak{h}$ is finite-dimensional. We now turn to the direct sum decomposition. Let $\boldsymbol{n} = (n_1, \dots, n_M) \in \{0, 1\}^M$ be an M-tuple of eigenvalues of the respective number operators, and define the joint projection as

$$P_{\boldsymbol{n}} \coloneqq \prod_{i=1}^{M} \left(n_i N_i + (1 - n_i)(\mathbb{1} - N_i) \right).$$

This is a product of commuting projections and is therefore still a projection, and still defined on the entire fermionic Fock space. The set $\{P_{\boldsymbol{n}} : \boldsymbol{n} \in \{0, 1\}^M\}$ is a complete set of projections, as shown by the completeness relation

$$\begin{aligned}
\sum_{\boldsymbol{n} \in \{0,1\}^M} P_{\boldsymbol{n}} &= \sum_{\boldsymbol{n}} \prod_{i=1}^{M} (n_i N_i + (1 - n_i)(\mathbb{1} - N_i)) \\
&= \prod_{i=1}^{M} \sum_{n_i=0}^{1} (n_i N_i + (1 - n_i)(\mathbb{1} - N_i)) \\
&= \prod_{i=1}^{M} \mathbb{1} = \mathbb{1},
\end{aligned}$$

again defined on the entire Fock space. We conclude by simply applying this completeness relation:

$$\mathcal{F}^{\mathrm{F}}(\mathfrak{h}) = \sum_{\boldsymbol{n}\in\{0,1\}^M} P_{\boldsymbol{n}}\mathcal{F}^{\mathrm{F}}(\mathfrak{h}) = \bigoplus_{\boldsymbol{n}\in\{0,1\}^M} P_{\boldsymbol{n}}\mathcal{F}^{\mathrm{F}}(\mathfrak{h}).$$

□

Remark 12. The case where $\mathfrak{h}$ is infinite-dimensional is not much different, and even then we can always choose a finite set of M independent number operators to obtain a decomposition into a direct sum of M (possibly infinite-dimensional) joint eigenspaces. But, as will be made clear later on, we will put a UV cutoff on our system and this will reduce the allowed momenta of a particle to a finite set. Having then a finite one-particle state space will make our life much easier, as we can make use of the finite basis given by the eigenvectors $\{|\boldsymbol{n}\rangle = |n_1\rangle \otimes \cdots \otimes |n_M\rangle : n_i \in \{0,1\}\ \forall i = 1,\ldots,M\}$, with eigenvalues given by the occupation of the M independent modes.

We can now come back to a statement that we previously used as motivation for the introduction of creation/annihilation operators, and show that indeed every observable in the grand canonical picture can be written in terms of such operators. We focus only on 1-body and 2-body observables because these are the types of operators that show up in the Hamiltonian we set out to study, but this procedure can be generalized to any N-body operator.

Theorem 1.10 (2nd quantization of 1-body operator)**.** *Let h be an observable on $\mathfrak{h}$ and $\{u_\alpha\}_{\alpha=1}^{\infty}$ be an orthonormal basis for $\mathfrak{h}$ with elements in the dense domain $D(h)$. Following the notation from the beginning of this section, we can write*

$$\bigoplus_{N=1}^{\infty}\sum_{j=1}^{N} h_j = \sum_{\alpha,\beta=1}^{\infty} \langle u_\alpha, hu_\beta\rangle_{\mathfrak{h}} a_\pm^*(u_\alpha)a_\pm(u_\beta)$$

on the domain $\cup_{M=0}^{\infty}\bigoplus_{N=1}^{M} P_\pm(D(\sum_{j=1}^{N} h_j))$ (for $M = 0$ the domain is taken to be $\mathbb{C}$). This operator is called the second quantization of h.

Proof. First we observe that the thesis holds in the sense of quadratic forms on $D(h)$. Indeed, let $f, g \in D(h)$ and recall the definition of creation/annihilation operators in 4

$$\begin{aligned}\sum_{\alpha,\beta=1}^{\infty} \langle u_\alpha, hu_\beta\rangle\langle g, a_\pm^*(u_\alpha)a_\pm(u_\beta)f\rangle &= \sum_{\alpha,\beta=1}^{\infty} \langle u_\alpha, hu_\beta\rangle\langle g, u_\alpha\rangle\langle u_\beta, f\rangle = \sum_{\beta=1}^{\infty}\langle g, hu_\beta\rangle\langle u_\beta, f\rangle \\ &= \sum_{\beta=1}^{\infty}\langle hg, u_\beta\rangle\langle u_\beta, f\rangle = \langle hg, f\rangle = \langle g, hf\rangle\end{aligned}$$

where the indices α and β are removed by using the completeness of the basis $\{u_\alpha\}_{\alpha=1}^{\infty}$, and h can be moved on either side of the inner products as it is a symmetric operator with $f, g \in D(h)$ and $u_\alpha \in D(h)$ for all α. Moving onto N-fold pure tensors, let $f_1, \ldots, f_N \in D(h)$ for some $N \geq 1$ and consider the action of $a^*(u_\alpha)a(u_\beta)$ on their tensor product

$$a^*(u_\alpha)a(u_\beta)f_1 \otimes \cdots \otimes f_N = N\langle u_\beta, f_1\rangle u_\alpha \otimes f_2 \otimes \cdots \otimes f_N$$

then it follows that

$$\sum_{\alpha,\beta=1}^{\infty} \langle u_\alpha, hu_\beta\rangle a^*(u_\alpha)a(u_\beta) = Nh_1$$

as quadratic form on finite linear combinations of pure tensor products of elements in $D(h)$. Recall the definition of the projections $P_\pm$ from equation 3, then it is clear that

$$P_\pm N h_1 P_\pm = P_\pm \sum_{j=1}^{N} h_j P_\pm = \sum_{j=1}^{N} h_j P_\pm$$

on $\bigotimes^N \mathfrak{h}$, as each h_j acts in the same way as h on the respective factor in the tensor product, which means that it maps the (anti-)symmetric subspace into itself. Then it follows that

$$\sum_{\alpha,\beta=1}^{\infty} \langle u_\alpha, h u_\beta\rangle a_\pm^*(u_\alpha) a_\pm(u_\beta) P_\pm = \sum_{\alpha,\beta=1}^{\infty} \langle u_\alpha, h u_\beta\rangle P_\pm a^*(u_\alpha) a(u_\beta) P_\pm = \bigoplus_{N=1}^{\infty} \sum_{j=1}^{N} h_j P_\pm$$

from which the thesis follows immediately. □

Remark 13. Another way to lift an operator U to the Fock space is multiplicatively as $\bigoplus_{N=0}^\infty \bigotimes^N U$, and this plays an important role in the discussion of symmetries. Given a Lie group (although this discussion can be somewhat generalized to groups that are not Lie groups) and a unitary representation γ, then $\mathrm{d}\gamma$ defines a representation of its associated Lie algebra as hermitian operators. One can then define a map Γ that lifts the representation γ multiplicatively to the Fock space, and this induces again another representation $\mathrm{d}\Gamma$ which now lifts the representation $\mathrm{d}\gamma$ additively to the Fock space. One can show that Γ indeed defines a representation, and this highlights how the two ways of lifting an operator to a Fock space respect two different algebraic structures (at least in this case where we discuss groups of symmetries of a system): a multiplicative (group) and a linear (algebra) structure. Given a unitary operator U on $\mathfrak{h}$ we can call $\Gamma(U)$ its multiplicative representation on the Fock space, and given a symmetric operator A we can call $\mathrm{d}\Gamma(A)$ its additive representation on the Fock space.

Remark 14. The second quantization of the identity operator on $\mathfrak{h}$ is the number operator

$$\mathcal{N} = \bigoplus_{N=0}^{\infty} \sum_{j=0}^{N} \mathbb{1}_j = \bigoplus_{N=0}^{\infty} N = \sum_{\alpha=1}^{\infty} a_\pm^*(u_\alpha) a_\pm(u_\beta).$$

Now it is clear why we called $a_\pm^*(f) a_\pm(f)$ the number *density* operator, as a number operator is the sum of operators that count the occupation number of a given state over a "complete set" of states. In a much more delicate way one could define creation/annihilation *distributions* as objects that formally create/annihilate states completely localized at one point in space. If we called $|x\rangle$ an "eigenstate" (in a generalized sense that we will not discuss here) of the position operator with eigenvalue x, then the number density would be written as $\rho(x) = a_\pm^*(|x\rangle) a_\pm(|x\rangle)$ and the total number of particles would turn into a very familiar expression: $\mathcal{N} = \int_{\mathbb{R}^n} \mathrm{d}x\, \rho(x)$, but to make sense of these objects we would need tools that go beyond the scope of our work.

Theorem 1.11 (2nd quantization of 2-body operator)**.** *Let W be a 2-body potential for identical particles, i.e. an observable on $\mathfrak{h} \otimes \mathfrak{h}$ such that* $\mathrm{ExWEx} = W$. *Let $\{u_\alpha\}_{\alpha=1}^\infty$ be an orthonormal basis for $\mathfrak{h}$ such that $u_\alpha \otimes u_\beta \in D(W)$ for all α and β. Then as a quadratic form on the domain $\cup_{M=2}^\infty \bigoplus_{N=2}^M P_\pm(D(\sum_{1\le i<j\le N} W_{ij}))$ we have*

$$\bigoplus_{N=2}^{\infty} \sum_{1\le i<j\le N} W_{ij} = \frac{1}{2} \sum_{\alpha,\beta,\nu,\mu=1}^{\infty} \langle u_\alpha \otimes u_\beta, W u_\mu \otimes u_\nu\rangle a_\pm^*(u_\alpha) a_\pm^*(u_\beta) a_\pm(u_\nu) a_\pm(u_\mu).$$

Again, this operator is called the second quantization of W. Note that in the fermionic case, the order in which the indices appear in the creation-creation and annihilation-annihilation products matters, as these operators anticommute rather than commute.

Proof. We proceed in a similar fashion to the second quantization of one-body operators. We have that

$$Wu_1 \otimes u_2 = \sum_{\alpha,\beta,\mu,\nu=1}^{\infty} \langle u_\alpha \otimes u_\beta, Wu_\mu \otimes u_\nu \rangle \langle u_\mu, u_1 \rangle \langle u_\nu, u_2 \rangle u_\alpha \otimes u_\beta,$$

using the resolution of the identity. But now notice that the product of creation/annihilation acts on pure tensors as

$$a^*(u_\alpha)a^*(u_\beta)a(u_\nu)a(u_\mu)u_1 \otimes \cdots \otimes u_N = N(N-1)\langle u_\mu, u_1 \rangle \langle u_\nu, u_2 \rangle u_\alpha \otimes u_\beta \otimes u_3 \otimes \cdots \otimes u_N,$$

and that, like in the one-body case, the projectors $P_\pm$ act on W as

$$\frac{N(N-1)}{2} P_\pm W_{12} P_\pm = \sum_{1 \le i < j \le N} W_{ij} P_\pm,$$

where we used that $\mathrm{Ex} W \mathrm{Ex} = W$. Recall, finally, that $P_\pm a^*(f) P_\pm = P_\pm a^*(f)$ as it doesn't matter if the starting state is (anti-)symmetrized since the image gets (anti-)symmetrized. With this in mind, we may write

$$\begin{aligned}
\bigoplus_{N=2}^{\infty} \sum_{1 \le i<j \le N} W_{ij} P_\pm &= \bigoplus_{N=2}^{\infty} \frac{N(N-1)}{2} P_\pm W_{12} P_\pm \\
&= P_\pm \frac{1}{2} \sum_{\alpha,\beta,\nu,\mu=1}^{\infty} \langle u_\alpha \otimes u_\beta, Wu_\mu \otimes u_\nu \rangle a^*(u_\alpha)a^*(u_\beta)a(u_\nu)a(u_\mu) P_\pm \\
&= \frac{1}{2} \sum_{\alpha,\beta,\nu,\mu=1}^{\infty} \langle u_\alpha \otimes u_\beta, Wu_\mu \otimes u_\nu \rangle P_\pm a^*(u_\alpha) P_\pm a^*(u_\beta)a(u_\nu)a(u_\mu) P_\pm \\
&= \frac{1}{2} \sum_{\alpha,\beta,\nu,\mu=1}^{\infty} \langle u_\alpha \otimes u_\beta, Wu_\mu \otimes u_\nu \rangle a_\pm^*(u_\alpha)a_\pm^*(u_\beta)a_\pm(u_\nu)a_\pm(u_\mu) P_\pm,
\end{aligned}$$

which concludes the proof. □

1.3 One-particle density matrices

It is clear from our discussion of the second quantization of an operator, that expectation values of any observable will inevitably reduce to expectation values of products of creation and annihilation operators. An important result that we will discuss in the section after this one states that – under appropriate conditions which are not at all obvious – the expectation of an observable can be written in terms of expectations of operators of the form $a_\pm^\# a_\pm^\#$. We therefore begin the study of these kinds of expectations, starting from the following definition.

Definition 1.15 (One-particle density matrix (1-pdm))**.** Let $\Psi = \bigoplus_{N=0}^{\infty} \Psi_N$ be a normalized vector on the bosonic or fermionic Fock space $\mathcal{F}^{\mathrm{B,F}}(\mathfrak{h})$ with finite particle number expectation

$$\langle \Psi, \mathcal{N}\Psi \rangle = \sum_{N=0}^{\infty} N \|\Psi_N\|^2 < \infty.$$

We define the *1-particle density matrix* or *2-point function* of Ψ as the operator γ_Ψ on $\mathfrak{h}$ given, as a quadratic form, by

$$\langle f, \gamma_\Psi g \rangle_{\mathfrak{h}} = \langle \Psi, a_\pm^*(g) a_\pm(f) \Psi \rangle$$

for all $f, g \in \mathfrak{h}$.

Proposition 1.12. *The operator γ_Ψ is positive semi-definite, i.e. $\gamma_\Psi \geq 0$, and trace class, with*

$$\mathrm{Tr}(\gamma_\Psi) = \langle \Psi, \mathcal{N}\Psi \rangle.$$

Proof. ??? $\square$

Lemma 1.13. *If Ψ is a finite linear combination of pure tensor products of elements from a subspace $X \subseteq \mathfrak{h}$ then γ_Ψ is a finite rank operator whose range is a subspace of X.*

Proof. ??? $\square$

Lemma 1.14. *If $\Psi \in \mathcal{F}^{\mathrm{F}}(\mathfrak{h})$ is a normalized vector then γ_Ψ satisfies the inequality*

$$0 \leq \gamma_\Psi \leq \mathbb{1}.$$

In particular, the eigenvalues of γ_Ψ lie in the interval $[0, 1]$.

Proof. We need to show that for every $f \in \mathfrak{h}$ it holds that $0 \leq \langle f, \gamma_\Psi f \rangle \leq \|f\|^2$. Using that $a(f)$ and $a^*(f)$ are adjoints of each other, we can see that

$$\langle f, \gamma_\Psi f \rangle = \langle \Psi, a_-^*(f) a_-(f) \Psi \rangle = \langle a_-(f)\Psi, a_-(f)\Psi \rangle = \|a_-(f)\Psi\|^2 \geq 0.$$

On the other hand, the CAR imply

$$\begin{aligned} \langle f, \gamma_\Psi f \rangle = \langle \Psi, a_-^*(f) a_-(f) \Psi \rangle &\leq \langle \Psi, a_-^*(f) a_-(f) \Psi \rangle + \langle a_-^*(f)\Psi, a_-^*(f)\Psi \rangle \\ &= \langle \Psi, (a_-^*(f) a_-(f) + a_-(f) a_-^*(f))\Psi \rangle = \langle \Psi, \{a_-^*(f), a_-(f)\}\Psi \rangle = \|f\|^2. \end{aligned}$$

This concludes the proof. $\square$

Lemma 1.15. *If $u_1, \ldots, u_N$ are orthonormal vectors in $\mathfrak{h}$, then the 1pdm of $\Psi = u_1 \wedge \cdots \wedge u_N$ is the projection onto the N-dimensional space* $\mathrm{span}\{u_1, \ldots, u_N\}$.

Proof. ??? $\square$

We are now ready to prove a fundamental theorem on which the treatment of any fermionic system is built, especially if perturbation theory is involved: the ground state energy and eigenvector of non-interacting fermions.

Theorem 1.16 (Ground state of non-interacting fermions)**.** *Consider the Hamiltonian $H_N^{\mathrm{in}} = \sum_{j=1}^N h_j$ for N particles, introduced in Equation 1, restricted to the anti-symmetric subspace $\bigwedge^N \mathfrak{h}$, i.e. with domain*

$$D_-(H_N^{\mathrm{in}}) = P_- D(H_N^{\mathrm{in}}).$$

Assume that the one-body operator h is bounded below, i.e. there exists some constant $E_0 \in \mathbb{R}$ such that $h \geq E_0$. The ground state energy of this fermionic system is

$$\inf \left\{ \mathrm{Tr}(Ph) : P \text{ is a projection onto an } N\text{-dimensional subspace of } D(h) \right\}.$$

If the infimum is attained for an N-dimensional projection P then H_N^{in} has a ground state eigenvector $f_1 \wedge \cdots \wedge f_N$, where $f_1, \ldots, f_N$ is an orthonormal basis for $P(\mathfrak{h})$. In particular, these vectors can be chosen to be eigenvectors of h, and the expectations of h restricted to $P(\mathfrak{h})^\perp \cap D(h)$ will be greater than the expectations of h restricted to $P(\mathfrak{h})$.

Proof. Since $D(h)$ is assumed to be dense, we can choose an orthonormal basis $\{u_\alpha\}_{\alpha=1}^\infty$ for $\mathfrak{h}$ of elements in $D(h)$. Let $\Psi \in D_-(H_N^{\text{in}})$ be a normalized vector, then

$$\langle \Psi, H_N^{\text{in}} \Psi \rangle = \sum_{\alpha,\beta=1}^{\infty} \langle u_\alpha, h u_\beta \rangle \langle \Psi, a_-^*(u_\alpha) a_-(u_\beta) \Psi \rangle = \sum_{\alpha,\beta=1}^{\infty} h_{\alpha\beta} \langle u_\beta, \gamma_\Psi u_\alpha \rangle.$$

Since, from the second quantization of an observable, Ψ is assumed to be a finite linear combination of pure tensor products of elements in $D(h)$, it follows from Lemma 1.13 that γ_Ψ has finite rank. Let $\{v_1, \ldots, v_n\}$ be the set of eigenvectors of γ_Ψ with eigenvalues $\{\lambda_1, \ldots, \lambda_n\}$, then it follows from the same lemma that $v_1, \ldots, v_n \in D(h)$. Then

$$\langle \Psi, H_N^{\text{in}} \Psi \rangle = \sum_{j=1}^{n} \sum_{\alpha,\beta=1}^{\infty} \lambda_j h_{\alpha\beta} \langle u_\beta, v_j \rangle \langle v_j, u_\alpha \rangle = \sum_{j=1}^{n} \lambda_j \langle v_j, h v_j \rangle.$$

Because $\Psi \in D_-(H_N^{\text{in}})$, it must have fixed particle number N, then it follows that

$$\sum_{j=1}^{n} \lambda_j = \operatorname{Tr}(\gamma_\Psi) = \langle \Psi, \mathcal{N} \Psi \rangle = N,$$

but $0 \le \lambda_j \le 1$, and therefore it must be that $n \ge N$. We may assume the eigenvectors v_j to be ordered such that $\langle v_1, h v_1 \rangle \le \ldots \le \langle v_n, h v_n \rangle$ so that, by defining P as the projection onto $\operatorname{span}\{v_1, \ldots, v_n\}$, we can provide the bound

$$\operatorname{Tr}(Ph) = \sum_{j=1}^{N} \langle v_j, h v_j \rangle \le \sum_{j=1}^{n} \lambda_j \langle v_j, h v_j \rangle = \langle \Psi, H_N^{\text{in}} \Psi \rangle.$$

Thus

$$\begin{aligned} &\inf\{\operatorname{Tr}(Ph) : P \text{ proj. onto N-dimensional subspace of } D(h)\} \\ &\qquad \le \inf\{\langle \Psi, H_N^{\text{in}} \Psi \rangle : \Psi \in D(H_N^{\text{in}}), \|\Psi\| = 1\}. \end{aligned}$$

It remains to show the opposite inequality. Let $\Psi = f_1 \wedge \cdots \wedge f_N$ with $f_1, \ldots, f_N \in D(h)$ orthonormal vectors. Again $\gamma_\Psi =: P$ is a projection onto $\operatorname{span}\{f_1, \ldots, f_N\}$, and as above we find that

$$\langle \Psi, H_N^{\text{in}} \Psi \rangle \le \operatorname{Tr}(\gamma_\Psi h)$$

and the inequality carries to the infimum, which finally proves the double inequality and the first part of the thesis. Assume now that P is a minimizing projection for the variational problem. It is clear that the state $\Psi = f_1 \wedge \cdots \wedge f_N$ whose 1pdm coincides with P is a ground state for H_N^{in} if $f_1, \ldots, f_N$ form an orthonormal basis for $P(\mathfrak{h})$.
We may now prove the stated properties for this minimizing projection. Let $\phi \in P(\mathfrak{h})$ and $\psi \in P(\mathfrak{h})^\perp \cap D(h)$ be two normalized states, then

$$\langle \phi, h\phi \rangle \le \langle \psi, h\psi \rangle.$$

Furthermore, consider the projection Q onto the N-dimensional space where ψ and ϕ are swapped, i.e. onto

$$\operatorname{span}(\{\psi\} \cup (P(\mathfrak{h}) \cap \{\phi\}^\perp)),$$

and since P is a minimizer it is clear that

$$0 \le \operatorname{Tr}(Qh) - \operatorname{Tr}(Ph) = \langle \psi, h\psi \rangle - \langle \phi, h\phi \rangle.$$

We can use this same argument to show that if $\psi \in D(h) \cap (P(\mathfrak{h}) \cap \{\phi\}^\perp)^\perp$ then $\langle \phi, h\phi \rangle \leq \langle \psi, h\psi \rangle$, and we now use this fact to show that h maps $P(\mathfrak{h})$ which leads to the conclusion that $P(\mathfrak{h})$ is indeed spanned by eigenvectors of h.
We proceed by contradiction, so assume that there exists a vector $\phi \in P(\mathfrak{h})$ such that $h\phi \notin P(\mathfrak{h})$. Since $D(h)$ is dense there is a $g \in D(h)$ such that $\langle (\mathbb{1} - P)g, h\phi \rangle = \langle g, (\mathbb{1} - P)h\phi \rangle \neq 0$, and, without loss of generality, we may assume $\mathrm{Re}\langle (\mathbb{1} - P)g, h\phi \rangle \neq 0$. We have that $(\mathbb{1} - P)g \in D(h)$. Consider $t \in \mathbb{R}$ (close to 0) and define

$$\phi_t = \frac{\phi + t(\mathbb{1} - P)g}{\|\phi + t(\mathbb{1} - P)g\|}.$$

Note that $\phi_t \in D(h) \cap (P(\mathfrak{h}) \cap \{\phi\}^\perp)^\perp$, and therefore $\langle \phi, h\phi \rangle \leq \langle \phi_t, h\phi_t \rangle$. But this is a contradiction, since

$$0 = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \langle \phi_t, h\phi_t \rangle = \langle (\mathbb{1} - P)g, h\phi \rangle + \langle \phi, h(\mathbb{1} - P)g \rangle = 2\mathrm{Re}\langle (\mathbb{1} - P)g, h\phi \rangle \neq 0.$$

□

Before going over the next section there is an important generalization to be made. Everything up to now relied on the possibility of writing the second quantization of any observable by means of products of creation and annihilation operators, once a basis for the one-particle Hilbert space was chosen. It is, however, quite natural to ask what happens - among other things - when a different basis is chosen, or when the Hamiltonian contains terms of the form $a^*(f)a^*(g)$ and $a(f)a(g)$. These, and other questions, can be answered by appropriately generalizing the one-particle density matrix, and the aim of this section is precisely to introduce this generalization. Our study case doesn't include terms of the above form, i.e. it only contains number-preserving interactions, but when going over Wick's theorem it will become quite clear just how powerful this generalization truly is, as it would eventually make it much easier to study the same problem through different physical lenses, using quasi-particle states, and also study much more general systems.

Definition 1.16 (Generalized 1pdm)**.** Given a normalized vector $\Psi \in \mathcal{F}^{\mathrm{F,B}}(\mathfrak{h})$ with finite particle number expectation, i.e. $\langle \Psi, \mathcal{N}\Psi \rangle < \infty$, we define the *generalized one-particle density matrix* as the positive semi-definite operator $\Gamma_\Psi : \mathfrak{h} \oplus \mathfrak{h}^* \to \mathfrak{h} \oplus \mathfrak{h}^*$ defined by

$$\langle f_1 \oplus Jg_1, \Gamma_\Psi f_2 \oplus Jg_2 \rangle_{\mathfrak{h} \oplus \mathfrak{h}^*} = \langle \Psi, (a_\pm^*(f_2) + a_\pm(g_2))(a_\pm(f_1) + a_\pm^*(g_1))\Psi \rangle_{\mathcal{F}^{\mathrm{F,B}}}$$

where J is the canonical isomorphism between $\mathfrak{h}$ and $\mathfrak{h}^*$.

It will be useful to recall that J acts as $Jg(f) = \langle g, f \rangle$, and its inverse is also its adjoint, i.e. $J^{-1} = J^*$. Given the normalized vector Ψ we may write Γ_Ψ in block-matrix form by using an auxiliary operator, $\alpha_\Psi : \mathfrak{h}^* \to \mathfrak{h}$, that "measures" how far the state Ψ is from having a fixed particle number. The operator is defined by

$$\langle f, \alpha_\Psi Jg \rangle = \langle \Psi, a_\pm(g)a_\pm(f)\Psi \rangle$$

for all $f, g \in \mathfrak{h}$. Note that this map has a useful property, which essentially relates its adjoint to the corresponding operator $a_\pm^*(f)a_\pm^*(g)$, that reads

$$\alpha_\Psi^* = \pm J\alpha_\Psi J,$$

as can be shown by noting that, using the CAR or the CCR,

$$\langle \alpha_\Psi^* f, Jg \rangle = \langle f, \alpha_\Psi Jg \rangle = \pm\langle g, \alpha_\Psi Jf \rangle = \pm\langle J\alpha_\Psi Jf, Jg \rangle.$$

With this operator we come to the following lemma.

Lemma 1.17. *Given a normalized state $\Psi \in \mathcal{F}^{\mathrm{F,B}}(\mathfrak{h})$ with finite particle number expectation, its generalized one-particle density matrix can be written in the block-matrix form as*

$$\Gamma_\Psi = \begin{pmatrix} \gamma_\Psi & \alpha_\Psi \\ \pm J\alpha_\Psi J & \mathbb{1} \pm J\gamma_\Psi J^* \end{pmatrix}$$

Proof. It suffices to prove that, as a quadratic form on $\mathfrak{h} \oplus \mathfrak{h}^*$, this matrix reproduces exactly Definition 1.16. Let $f_1, g_1, f_2, g_2 \in \mathfrak{h}$, then it holds that

$$\begin{aligned}
&\begin{pmatrix} f_1 & Jg_1 \end{pmatrix} \begin{pmatrix} \gamma_\Psi & \alpha_\Psi \\ \pm J\alpha_\Psi J & \mathbb{1} \pm J\gamma_\Psi J^* \end{pmatrix} \begin{pmatrix} f_2 \\ Jg_2 \end{pmatrix} = \begin{pmatrix} f_1 & Jg_1 \end{pmatrix} \begin{pmatrix} \gamma_\Psi f_2 + \alpha_\Psi J g_2 \\ \pm J\alpha_\Psi J f_2 + (\mathbb{1} \pm J\gamma_\Psi J^*) J g_2 \end{pmatrix} \\
&= \langle f_1, \gamma_\Psi f_2 \rangle + \langle f_1, \alpha_\Psi J g_2 \rangle + \langle + \langle J g_1, \alpha_\Psi^* f_2 \rangle + \langle J g_1, (\mathbb{1} \pm J\gamma_\Psi J^*) J g_2 \rangle \\
&= \langle a_\pm^*(f_2) a_\pm(f_1) \rangle + \langle a_\pm(g_2) a_\pm(f_1) \rangle + \langle a_\pm^*(f_2) a_\pm^*(g_1) \rangle + \langle a_\pm(g_2) a_\pm^*(g_1) \rangle \\
&= \langle (a_\pm^*(f_2) + a_\pm(g_2))(a_\pm(f_1) + a_\pm^*(g_1)) \rangle \\
&= \langle f_1 \oplus J g_1, \Gamma_\Psi f_2 \oplus J g_2 \rangle.
\end{aligned}$$

□

The usefulness of this new object is that it allows for a very simple way to write this quantity in a way that closely resembles the standard one-particle density matrix. We just need to define the generalized creation/annihilation operators:

Definition 1.17. Let $f, g \in \mathfrak{h}$ be one-particle states. The generalized annihilation and creation operators are defined, respectively, by the maps

$$\begin{aligned} A_\pm(f \oplus Jg) &\coloneqq a_\pm(f) + a_\pm^*(g) \\ A_\pm^*(f \oplus Jg) &\coloneqq a_\pm^*(f) + a_\pm(g). \end{aligned}$$

Remark 15. Once again we have that the generalized annihilation operators are conjugate linear from $\mathfrak{h} \oplus \mathfrak{h}^*$ to operators on $\mathcal{F}^{\mathrm{F,B}}(\mathfrak{h})$. Furthermore, we have the following relation between these generalized operators. For all $F \in \mathfrak{h} \oplus \mathfrak{h}^*$ it holds that

$$A_\pm^*(F) = A_\pm(\mathcal{J}F), \quad \text{where } \mathcal{J} = \begin{pmatrix} 0 & J^* \\ J & 0 \end{pmatrix} : \mathfrak{h} \oplus \mathfrak{h}^* \to \mathfrak{h} \oplus \mathfrak{h}^*$$

Proposition 1.18 (Generalized CCR and CAR)**.** *For all $F_1, F_2 \in \mathfrak{h} \oplus \mathfrak{h}^*$ we have the following generalized versions of the canonical (anti-)commutation relations. For bosons*

$$[A_+(F_1), A_+^*(F_2)] = \langle F_1, \mathcal{S} F_2 \rangle_{\mathfrak{h} \oplus \mathfrak{h}^*} \quad \text{where } \mathcal{S} = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix} : \mathfrak{h} \oplus \mathfrak{h}^* \to \mathfrak{h} \oplus \mathfrak{h}^*,$$

and for fermions

$$\{A_-(F_1), A_-^*(F_2)\} = \langle F_1, F_2 \rangle_{\mathfrak{h} \oplus \mathfrak{h}^*}.$$

More compactly we could introduce the natural notation $[A_\pm(F_1), A_\pm^*(F_2)]_\mp = \langle F_1, \mathcal{S}_\pm F_2 \rangle$, where we define $\mathcal{S}_-$ as simply being the identity, and the sign on the square brackets indicates a commutator (-) or an anticommutator (+).

Proof. The proof is trivial, as it only requires to expand the expression for the (anti-)commutator and to use explicitly the CCR/CAR. □

Remark 16. Note that, in general, the (anti-)commutators $[A_\pm(F), A_\pm(G)]_\mp$ and $[A_\pm^*(F), A_\pm^*(G)]_\mp$ need not be zero.

Remark 17. The special case where $F = f \oplus 0$ reads $A_\pm^\#(F) = a_\pm^\#(f)$.

With these operators, the generalized one-particle density matrix can be written in a familiar form. Given, as usual, $\Psi \in \mathcal{F}^{\mathrm{F,B}}(\mathfrak{h})$ normalized and for all $F_1, F_2 \in \mathfrak{h} \oplus \mathfrak{h}^*$ we have

$$\langle F_1, \Gamma_\Psi F_2 \rangle_{\mathfrak{h} \oplus \mathfrak{h}^*} = \langle \Psi, A_\pm^*(F_1) A_\pm(F_2) \Psi \rangle$$

1.4 Bogolubov transformations and quasi-free pure states

We can finally realize the notion of a "change of basis" for creation/annihilation operators on the space $\mathfrak{h} \oplus \mathfrak{h}^*$. The following definition characterizes a class of isomorphisms on this space as those that, in some sense, preserve the structure of the generalized creation and annihilation operators.

Definition 1.18 (Bogolubov transformations)**.** A linear bounded isomorphism $\mathcal{V} : \mathfrak{h} \oplus \mathfrak{h}^* \to \mathfrak{h} \oplus \mathfrak{h}^*$ is called a (bosonic or fermionic) *Bogolubov map* if, for all $F, G \in \mathfrak{h} \oplus \mathfrak{h}^*$

- $A^*(\mathcal{V}F) = A(\mathcal{V}\mathcal{J}F)$;
- $[A_\pm(\mathcal{V}F), A_\pm^*(\mathcal{V}G)]_\mp = \langle F, \mathcal{S}_\pm G\rangle$.

One also refers to the map $A_\pm(F) \longmapsto A_\pm(\mathcal{V}F)$ as a *Bogolubov transformation.*

We can characterize Bogolubov maps by writing them in block-matrix form and noting that each block must satisfy specific properties that come directly from the definition. We first notice that the conditions in the definition can be rewritten as

$$\langle \mathcal{V}F_1, \mathcal{S}_\pm \mathcal{V}F_2\rangle = \langle F_1, \mathcal{S}_\pm F_2\rangle \quad \text{and} \quad \mathcal{J}\mathcal{V}F = \mathcal{V}\mathcal{J}F$$

for all $F, F_1, F_2 \in \mathfrak{h} \oplus \mathfrak{h}^*$. Moreover, since we are assuming that $\mathcal{V}$ is invertible, we can also write

$$\mathcal{V}^{-1} = \mathcal{S}_\pm \mathcal{V}^* \mathcal{S}_\pm.$$

In particular, $\mathcal{V}^{-1} = \mathcal{V}^*$ in the fermionic case. With these properties we immediatly get to the following proposition, for which the proof can be read through what we just said:

Proposition 1.19. *A linear map $\mathcal{V}$ on $\mathfrak{h} \oplus \mathfrak{h}^*$ is a Bogolubov map if, and only if, the following conditions hold:*

$$\mathcal{V}^*\mathcal{S}_\pm\mathcal{V} = \mathcal{S}_\pm, \quad \mathcal{V}\mathcal{S}_\pm\mathcal{V}^* = \mathcal{S}_\pm, \quad \mathcal{J}\mathcal{V}\mathcal{J} = \mathcal{V}. \tag{5}$$

If we take bounded linear maps $U : \mathfrak{h} \to \mathfrak{h}$ and $V : \mathfrak{h} \to \mathfrak{h}^*$ we can finally write a Bogolubov map in block-matrix form

$$\mathcal{V} = \begin{pmatrix} U & J^*VJ^* \\ V & JUJ^* \end{pmatrix} \tag{6}$$

and the conditions 5 can then be read as

$$\begin{cases} U^*U = \mathbb{1} \pm V^*V \\ UU^* = \mathbb{1} \pm J^*vV^*J \\ JV^*JU = \pm JU^*J^*V \end{cases}.$$

Defining Bogolubov maps is not sufficient though, as such a map need not, in general, correspond to a unitary *on the Fock space.* Thus we come to the following theorem which tells us when this is possible, and which we will use to define an important class of states in a Fock space for which Wick's theorem will hold in its most general form.

Theorem 1.20 (Unitary implementability of Bogolubov maps)**.** *If $\mathcal{V} : \mathfrak{h}\oplus\mathfrak{h}^* \to \mathfrak{h}\oplus\mathfrak{h}^*$ is a Bogolubov map of the form 6 then there exists a unitary map*

$$\mathbb{U}_\mathcal{V} : \mathcal{F}^{\mathrm{F,B}}(\mathfrak{h}) \to \mathcal{F}^{\mathrm{F,B}}(\mathfrak{h})$$

such that, for $F \in \mathfrak{h} \oplus \mathfrak{h}^*$,

$$\mathbb{U}_\mathcal{V} A_\pm(F) \mathbb{U}_\mathcal{V}^* = A_\pm(\mathcal{V}F)$$

if, and only if, V^*V is trace-class. This trace-class condition is also referred to as the Shale-Stinespring condition.

What this means is that, under suitable conditions on the block-matrix form of a Bogolubov map, there exists a "change of basis" on the Fock space that sends generalized annihilation and - by taking the adjoint - creation operators to operators that annihilate/create a superposition of states determined by the linear map $\mathcal{V}$.

Proof. AAAAAAHHHHHHH $\square$

We are finally ready to discuss the most important concept in this chapter: Wick's theorem. This is the result that, among other things, allows us to write perturbative series as they are described in the next chapter, and it is motivation behind the computational tools we will need to introduce. Thanks to the formalism of second quantization, Wick's theorem essentially states that the expectation value for any observable on a special class of states is completely determined by the 2-point function. Moreover, since the perturbative expansion of a physical quantity involves a series of products of creation/annihilation operators, this theorem also says that such a perturbative expansion is completely determined by the 2-point function as well. We first give the definition for pure states and then move onto the more important case of mixed states and prove their version of Wick's theorem.

Definition 1.19 (Quasi-free pure state)**.** A vector $\Psi \in \mathcal{F}^{\mathrm{F,B}}(\mathfrak{h})$ is called a *quasi-free pure state* if there exists a unitarily implementable Bogolubov map $\mathcal{V}$ such that $\Psi = \mathbb{U}_{\mathcal{V}}|0\rangle$, where $\mathbb{U}_{\mathcal{V}}$ is the unitary implementation of $\mathcal{V}$.

That is, a pure state is quasi-free if it is the vacuum state under an appropriate choice of physical modes.

Theorem 1.21 (Wick's theorem)**.** *If Ψ is a quasi-free pure state and $F_1, \dots, F_{2m} \in \mathfrak{h} \oplus \mathfrak{h}^*$ for some $m \geq 1$, then it holds that*

$$\begin{aligned}&\langle \Psi, A_\pm(F_1) \cdots A_\pm(F_2 m)\Psi\rangle = \\ &\quad \sum_{\sigma \in P_{2m}} (\pm 1)^\sigma \langle \Psi, A_\pm(F_{\sigma(1)}) A_\pm(F_{\sigma(2)})\Psi\rangle \cdots \langle \Psi, A_\pm(F_{\sigma(2m-1)}) A_\pm(F_{\sigma(2m)})\Psi\rangle\end{aligned} \tag{7}$$

and

$$\langle \Psi, A_\pm(F_1) \cdots A_\pm(F_{2m-1})\Psi\rangle = 0$$

where P_{2m} is the set of pairings of the indices $\{1, \dots, 2m\}$, i.e.

$$\begin{aligned}P_{2m} = \{\sigma \in S_{2m} : \sigma(2j-1) < \sigma(2j+1),\ j = 1, \dots, m-1 \text{ and} \\ \sigma(2j-1) < \sigma(2j),\ j = 1, \dots, m\}.\end{aligned}$$

Despite dreaming of resisting the urge to characterize *something*, the mathematician is ultimately doomed to fail. Let us then indulge in the characterization of generalized 2-point functions of quasi-free pure states, since they appear to be so important.

Proposition 1.22. *If $\Psi \in \mathcal{F}^{\mathrm{F,B}}(\mathfrak{h})$ is a quasi-free pure state then the generalized 2-point function, $\Gamma = \Gamma_\Psi$, satisfies the condition*

$$\Gamma \mathcal{S}_\pm \Gamma = \mp \Gamma. \tag{8}$$

In particular, Γ is a projection in the fermionic case. Conversely, if $\Gamma : \mathfrak{h} \oplus \mathfrak{h}^* \to \mathfrak{h} \oplus \mathfrak{h}^*$ is a positive semi-definite operator satisfying 8 and of the form

$$\Gamma = \begin{pmatrix} \gamma & \alpha \\ \pm J\alpha J & \mathbb{1} \pm J\gamma J^* \end{pmatrix}$$

with γ a trace-class operator, then there exists a quasi-free pure state $\Psi \in \mathcal{F}^{\mathrm{F,B}}(\mathfrak{h})$ such that $\Gamma_\Psi = \Gamma$.

We may now extend the definition of quasi-free pure states to include non-pure ones, and we may as well use Wick's theorem as the defining property of quasi-free expectations. Surprisingly enough, Gibbs states are quasi-free, as illustrated by the theorem we are about to prove.

Definition 1.20 (General quasi-free states)**.** A *quasi-free state* is a quantum state $\rho \in \mathfrak{B}^*(\mathcal{F}^{\mathrm{F,B}}(\mathfrak{h}))$ that satisfies Wick's theorem, i.e. such that Equation (7) holds when replacing expectations $\langle \Psi, \cdot \Psi \rangle$ with mixed expectations $\langle \cdot \rangle_\rho$.

Theorem 1.23. *Let $\{F_n\}_{n\in\mathbb{N}}$ be an orthonormal basis for $\mathfrak{h} \oplus \mathfrak{h}^*$ and define the Hamiltonian $H = \sum_{n=0}^{\infty} \epsilon_n A_\pm^*(F_n) A_\pm(F_n)$, where $\mathrm{Tr}(e^{-\beta H}) = \sum_n e^{-\beta \epsilon_n} < \infty$. The Gibbs state $\langle \cdot \rangle_{\beta,H}$ is quasi-free.*

Proof. LOOK AT GIULIANI AND VIGNALE. $\square$

1.5 Quadratic hamiltonians

Having introduced all the necessary results reguarding Bogolubov maps, we can now move onto the study of generic quadratic Hamiltonians. Their importance is partly due to the fact that if one can find a Bogolubov transformation that diagonalizes the quadratic part of the Hamiltonian, then the study of the kinetic (quadratic) part of that system is automatically done. This of course plays a crucial role in perturbative calculations, which are all about dealing with complicated interactions on top of an easy and well-defined “free theory”, i.e. a theory whose n-point functions satisfy Wick's theorem.

Definition 1.21 (Quadratic Hamiltonian)**.** Let $\mathcal{A} : \mathfrak{h} \oplus \mathfrak{h}^* \to \mathfrak{h} \oplus \mathfrak{h}^*$ be a self-adjoint, trace-class operator and assume morover in the bosonic case that it is positive semi-definite. Given an orthonormal basis $\{F_1, \ldots, F_{2M}\}$ for $\mathfrak{h} \oplus \mathfrak{h}^*$, the operator

$$H_{\mathcal{A}}^{\pm} = \sum_{i,j=1}^{2M} \langle F_i, \mathcal{A} F_j \rangle A_\pm^*(F_i) A_\pm(F_j),$$

is called the *quadratic Hamiltonian associated to $\mathcal{A}$*.

2 Perturbation Theory

2.1 A toy model

To compute expectation values of products of field operators, one can follow the Feynman integral approach. The aim is to define this integral without trying to integrate over field configurations, and in some cases this can be done rigorously by defining this special integral as a linear operator on commuting (bosonic) or anticommuting (fermionic) variables in such a way that the integral of polynomials respects Wick's rule. Bosonic field operators can be represented with complex variables, this way their expectation values become complex Gaussian integrals. Fermionic field operators on the other hand require a different algebraic structure.

2.2 Bosonic expectations

2.3 Grassmann Algebras and Grassmann Integrals

Consider the unital complex Grassmann algebra generated by a set of $2N$ anticommuting vectors:

$$\begin{aligned} &\mathcal{G}_N = \mathrm{span}\{1, \psi_1, \bar{\psi}_1, \ldots, \psi_N, \bar{\psi}_N\} \\ &\{\psi_i, \psi_j\} = 0 \quad \forall i, j = 1, ..., N \\ &\{\psi_i, \bar{\psi}_j\} = 0 \quad \forall i, j = 1, ..., N \\ &\{\bar{\psi}_i, \bar{\psi}_j\} = 0 \quad \forall i, j = 1, ..., N \end{aligned} \tag{9}$$

where $\{\psi, \phi\} := \psi\phi + \phi\psi$ is the anticommutator. The idea is to interpret the elements of this algebra as (polynomials of) field operators, so we need to define integrals and expectations of Grassmann variables, i.e. of elements of this algebra.
Given $\mathcal{G}_N$ the Grassmann algebra generated by $\{\psi_1, ..., \psi_N, \bar{\psi}_1, ..., \bar{\psi}_N\}$, denote by

$$\mathcal{G}_N^- = \{\phi \in \mathcal{G}_N : [\phi, \psi] = 0 \ \forall \psi \in \mathcal{G}_N\}$$
$$\mathcal{G}_N^+ = \{\phi \in \mathcal{G}_N : \{\phi, \psi\} = 0 \ \forall \psi \in \mathcal{G}_N\}$$

the even (commuting) and odd (anticommuting) subsets of the algebra. It is easy to show that these are subspaces (by linearity of commutators and anticommutators), and one can also show that $\mathcal{G}_N$ can be decomposed as

$$\mathcal{G}_N \cong \mathcal{G}_N^+ \oplus \mathcal{G}_N^-.$$

Odd elements satisfy a really important property. If $\psi \in \mathcal{G}_N^+$ it follows that $\{\psi, \psi\} = 2\psi^2 = 0$. That is, all powers greater than one of an odd element vanish.

Definition 2.1. Given $\mathcal{G}$ a Grassmann algebra, as defined above, and $f : \mathbb{C}^k \to \mathbb{C}$ an analytic function given by its Taylor series centered at zero

$$f(\boldsymbol{z}) = \sum_{j=0}^{\infty} \sum_{|\alpha|=j} \frac{\partial^\alpha f(0)}{j!} \boldsymbol{z}^i$$

we define the associated *function of Grassmann variables* as

$$f(\boldsymbol{\psi}) = \sum_{j=0}^{\infty} \sum_{|\alpha|=j} \frac{\partial^\alpha f(0)}{j!} \boldsymbol{\psi}^i$$

where $\boldsymbol{\psi} \in \mathcal{G}_N \times ... \times \mathcal{G}_N$ is the list of k Grassmann variables.

Remark 18. If $\psi \in \mathcal{G}_N^+$ it follows that any function of Grassmann variables evaluated in ψ has at most first order terms, as all higher orders will always contain a factor of $\psi^2 = 0$. The most important example to us is the exponential of an odd element:

$$e^\psi = 1 + \psi + \frac{1}{2}\psi^2 + ... = 1 + \psi$$

Remark 19. We could define a Grassmann algebra more "directly" as the ring of polynomials in anticommuting variables over a field, and choose $\mathbb{C}$ as base field to get exactly the structure we need. Indeed, the Grassmann algebra can be viewed as the free algebra generated by the variables corresponding to the modes of the physical system, quotiented by the ideal generated by anticommutators, i.e.

$$\mathcal{G}_N := \frac{\mathbb{C}\langle \psi_1, ..., \psi_N, \bar{\psi}_1, ..., \bar{\psi}_N \rangle}{\operatorname{span}\{\psi_i\psi_j + \psi_j\psi_i,\, \psi_i\bar{\psi}_j + \bar{\psi}_j\psi_i,\, \bar{\psi}_i\bar{\psi}_j + \bar{\psi}_j\bar{\psi}_i : i, j = 1, ..., N\}}$$

We now introduce integrals of functions of Grassmann variables. Note that the definition is going to be purely algebraic and, in particular, motivated by our need to reproduce specific computational rules. Indeed, we'll see that we can define expectations of functions of Grassmann variables in a way that reproduces exactly Wick's rule for expectations of fermionic field operators.
We ask, first of all, for the integral to be a linear operator on a Grassmann algebra. Then, specifying the image of a basis for the algebra will uniquely determine the operator.

Definition 2.2. Given $\{1, \psi_1, ..., \psi_N, \bar{\psi}_1, ..., \bar{\psi}_N\}$ the basis for the Grassmann algebra, as above, the *Grassmann integral* is the linear operator that acts on this basis as

$$\int \mathrm{d}\psi_i \, \psi_j = \int \mathrm{d}\bar{\psi}_i \, \bar{\psi}_j := \delta_{ij},$$
$$\int \mathrm{d}\psi_i = \int \mathrm{d}\bar{\psi}_i := 0.$$

Repeated integration is denoted by a single integral symbol and multiple "differentials".

Example 1. In $\mathcal{G}_2$, consider the exponential function. Using the definition, we can evaluate the following integrals to see how this operator behaves.

- $\int \mathrm{d}\psi_1 \, e^{\psi_1 \bar{\psi}_2} = \int \mathrm{d}\psi_1 \, (1 + \psi_1 \bar{\psi}_2) = \bar{\psi}_2;$
- $\int \mathrm{d}\bar{\psi}_2 \, e^{\psi_1 \bar{\psi}_2} = \int \mathrm{d}\bar{\psi}_2 \, (1 - \bar{\psi}_2 \psi_1) = -\psi_1;$
- $\int \mathrm{d}\psi_1 \mathrm{d}\bar{\psi}_2 \, e^{\psi_1 \bar{\psi}_2} = -1.$

Remark 20. Repeated integration, i.e. the composition of multiple integral operators, inherits some form of anticommutativity when acting on the basis elements. Indeed in $\mathcal{G}_1$, for example, we have

$$\begin{aligned} \int \mathrm{d}\psi \mathrm{d}\bar{\psi} \, \psi\bar{\psi} &= -\int \mathrm{d}\psi \mathrm{d}\bar{\psi} \, \bar{\psi}\psi = -1 \\ &= -\int \mathrm{d}\bar{\psi} \mathrm{d}\psi \, \psi\bar{\psi}. \end{aligned}$$

And given the definition of Grassmann functions, this property will be quite useful to us.

Having defined integration, we can now introduce a *Gaussian measure* (in a purely algebraic sense). To make the notation ligher, we define a "dot product", without any geometric meaning, as $\boldsymbol{\psi} \cdot \boldsymbol{\phi} := \sum_{i=1}^{k} \psi_i \phi_i$.

Definition 2.3. $G \in \mathrm{Mat}(N \times N, \mathbb{C})$ is said to be a *Grassmann covariance* if $G \in \mathrm{GL}(N, \mathbb{C})$.

Lemma 2.1. *Let G be a Grassmann covariance, $\boldsymbol{\psi} = (\psi_1, \ldots, \psi_N)$ and $\bar{\boldsymbol{\psi}} = (\bar{\psi}_1, \ldots, \bar{\psi}_N)$. Then*

$$\int \left(\prod_{i=1}^{N} \mathrm{d}\bar{\psi}_i \mathrm{d}\psi_i \right) e^{-\bar{\boldsymbol{\psi}} \cdot G^{-1} \boldsymbol{\psi}} = \det(G^{-1})$$

Proof. We carry out the calculation explicitly.

$$\begin{aligned} \int \left(\prod_{i=1}^{N} \mathrm{d}\bar{\psi}_i \mathrm{d}\psi_i \right) e^{-\bar{\boldsymbol{\psi}} \cdot G^{-1} \boldsymbol{\psi}} &= \int \left(\prod_{i=1}^{N} \mathrm{d}\bar{\psi}_i \mathrm{d}\psi_i \right) \frac{(-1)^N}{N!} \left(\sum_{i,j=1}^{N} \bar{\psi}_i (G^{-1})_{ij} \psi_j \right)^N \\ &= \int \left(\prod_{i=1}^{N} \mathrm{d}\bar{\psi}_i \mathrm{d}\psi_i \right) \frac{1}{N!} \left(\sum_{i,j=1}^{N} \bar{\psi}_i (G^{-1})_{ij} \psi_j \right)^N \\ &= \int \left(\prod_{i=1}^{N} \mathrm{d}\bar{\psi}_i \mathrm{d}\psi_i \right) \frac{1}{N!} \sum_{\sigma,\tau \in \mathrm{S}_N} \prod_{j=1}^{N} (G^{-1})_{\sigma(j),\tau(j)} \bar{\psi}_{\sigma(j)} \psi_{\tau(j)} = \star \end{aligned}$$

Where S_N is the set of permutations of N elements. We can split the product into a product of matrix elements times a product of Grassmann variables, then we use that

$$\prod_{j=1}^{N} \bar{\psi}_{\sigma(j)} \psi_{\tau(j)} = (-1)^{\mathrm{sgn}(\sigma) + \mathrm{sgn}(\tau)} \prod_{j=1}^{N} \bar{\psi}_j \psi_j$$

to find that

$$\begin{aligned}
\star &= \frac{1}{N!}\sum_{\sigma,\tau\in \mathrm{S}_N}(-1)^{\mathrm{sgn}(\sigma)+\mathrm{sgn}(\tau)}\prod_{j=1}^{N}(G^{-1})_{\sigma(j),\tau(j)}\\
&= \frac{1}{N!}\sum_{\sigma\in \mathrm{S}_N}(-1)^{\mathrm{sgn}(\sigma)}\prod_{j=1}^{N}(G^{-1})_{j,\sigma j}\\
&= \det(G^{-1})
\end{aligned}$$

□

Definition 2.4. Let G be a Gaussian covariance, and $\{\psi_1,\dots,\psi_N,\bar\psi_1,\dots,\bar\psi_N\}$ the usual Grassmann variables. A *Grassmann Gaussian measure* is the Grassmann integral measure defined by

$$P_G(\mathrm{d}\psi) := \frac{1}{\det(G^{-1})}\left(\prod_{j=1}^{N}\mathrm{d}\bar\psi_j\mathrm{d}\psi_j\right)e^{-\bar{\boldsymbol{\psi}}\cdot G^{-1}\boldsymbol{\psi}}$$

Remark 21. The factor in front of the differentials guarantees a "normalized" measure, i.e. $\int P_G(\mathrm{d}\psi) = (1/\det(G^{-1}))\det(G^{-1}) = 1$.

We will prove the following statement later.

Lemma 2.2. *With the same notation used so far,*

$$\int P_G(\mathrm{d}\psi)\psi_i\bar\psi_j = G_{ij}$$

Lemma 2.3. *Let $D\in \mathrm{GL}(2,\mathbb{C})$, $\boldsymbol{\varphi}$ and $\boldsymbol{\eta}$ two vectors of Grassmann variables. Then, under the following change of variables*

$$\begin{pmatrix}\boldsymbol{\psi}\\ \bar{\boldsymbol{\psi}}\end{pmatrix} = D\begin{pmatrix}\boldsymbol{\varphi}\\ \bar{\boldsymbol{\varphi}}\end{pmatrix} + \begin{pmatrix}\boldsymbol{\eta}\\ \bar{\boldsymbol{\eta}}\end{pmatrix}$$

the integral of a function of Grassmann variables f changes as follows

$$\int\left(\prod_i \mathrm{d}\bar\psi_i\psi_i\right)f(\boldsymbol{\psi},\bar{\boldsymbol{\psi}}) = \det(G^{-1})\int\left(\prod_i \mathrm{d}\bar\varphi_i\varphi_i\right)f(D(\boldsymbol{\varphi},\bar{\boldsymbol{\varphi}})+(\boldsymbol{\eta},\bar{\boldsymbol{\eta}}))$$

Proof. check proof in the notes (is D defined only in the diagonal NxN blocks? How are the "differentials" of the new variables calculated ?). □

Lemma 2.4. *Let $G\in \mathrm{GL}(N,\mathbb{C})$, then*

$$\int P_G(\mathrm{d}\psi)\,e^{\bar{\boldsymbol{\eta}}\cdot\boldsymbol{\psi}+\bar{\boldsymbol{\psi}}\cdot\boldsymbol{\eta}} = e^{\bar{\boldsymbol{\eta}}\cdot G\boldsymbol{\eta}}$$

Proof. it was an exercise. □

Before defining expectations, we need one more operator on a Grassmann algebra to mimic the behaviour of these same integrals done in the complex variables case. That is, we need to introduce derivatives of Grassmann functions. If we had a polynomial of degree N in N variables, then deriving this polynomial in all of the variables should return the maximal degree term. Then this becomes a natural choice for the definition of Grassmann derivatives, as we defined functions to be nothing more than Taylor series, and evaluating a function in the elements of the basis gets us exactly in this situation. But recall that integrating a Grassmann function in all variables (i.e., in all the basis elements) returns exactly the same thing, the coefficient of the maximal degree term. Therefore, this natural definition for Grassmann derivatives reads as follows.

Definition 2.5. Let $f(\boldsymbol{\psi}, \bar{\boldsymbol{\psi}})$ be a Grassmann function, the *Grassmann derivatives* of f w.r.t. the basis elements are the linear operators defined as

$$\frac{\partial}{\partial \psi_i} f := \int \mathrm{d}\psi_i \, f(\boldsymbol{\psi}, \bar{\boldsymbol{\psi}})$$

Proof (Lemma 2.2). Recalling the "anticommutativity property" of Grassmann integrals, it is clear that the same property holds for derivatives too. We can make use of Grassmann derivatives to prove this lemma in a way that is closer to the complex variables case, and this motivates the introduction of Grassmann derivatives. Note that, given a vector of Grassmann variables $\boldsymbol{\eta}$,

$$\psi_i \bar{\psi}_j e^{-\bar{\boldsymbol{\psi}} \cdot G^{-1} \boldsymbol{\psi}} = -\frac{\partial}{\partial \bar{\eta}_i} \frac{\partial}{\partial \eta_j} e^{-\bar{\boldsymbol{\psi}} \cdot G^{-1} \boldsymbol{\psi} + \bar{\boldsymbol{\eta}} \cdot \boldsymbol{\psi} + \bar{\boldsymbol{\psi}} \cdot \boldsymbol{\eta}} \bigg|_{\boldsymbol{\eta} = \bar{\boldsymbol{\eta}} = 0}.$$

We can substitute this in the integral to get that

$$\begin{aligned}
\int P_G(\mathrm{d}\psi) \, \psi_i \bar{\psi}_j &= -\frac{\partial}{\partial \bar{\eta}_i} \frac{\partial}{\partial \eta_j} \int P_G(\mathrm{d}\psi) e^{\bar{\boldsymbol{\eta}} \cdot \boldsymbol{\psi} + \bar{\boldsymbol{\psi}} \cdot \boldsymbol{\eta}} \bigg|_0 \\
&= -\frac{\partial}{\partial \bar{\eta}_i} \frac{\partial}{\partial \eta_j} e^{\bar{\boldsymbol{\eta}} \cdot G \boldsymbol{\eta}} \bigg|_0 \\
&= -\frac{\partial}{\partial \bar{\eta}_i} \frac{\partial}{\partial \eta_j} (\bar{\eta}_i G_{ij} \eta_j) \bigg|_0 \\
&= G_{ij}
\end{aligned}$$

which is exactly the thesis. □

With this useful lemma proven, we get to the most important theorem of this section: Wick's theorem. It is this result that allows us to actually interpret Grassmann Gaussian integrals as "expectations of fermionic field operators", because it shows that these integrals exactly reproduce the combinatorics of fermionic expectations.

Theorem 2.5 (Wick's rule)**.** *Let $M \in \mathbb{N}$ be the degree of a monomial in Grassmann variables, and let $I = \{a_1, \ldots, a_M\}$ and $J = \{b_1, \ldots, b_M\}$ be two sets of M indices. The integral of a monomial is*

$$\int P_G(\mathrm{d}\psi) \left(\prod_{\alpha \in I, \beta \in J} \psi_\alpha \bar{\psi}_\beta \right) = \sum_{\sigma \in \mathrm{S}_N} (-1)^{\mathrm{sgn}(\sigma)} \prod_{j=1}^{M} G_{a_j, b_{\sigma(j)}}$$

Proof. check proof of Isserlis' theorem, and check the way the product on the LHS is written. □

I REWORKED THE STRUCTURE OF THIS SECOND CHAPTER!. It follows that calling a Grassmann Gaussian integral a "fermionic expectation" makes sense. Our task now is to give important results that greatly simplify the calculation of such expectations. In particular, the aim is to represent these calculations using Feynman diagrams, which are a tool to perform the combinatorics of this exact type of problem, and because these integrals are supposed to be a representation of expectations of physical quantities, Feynman diagrams will have a clear physical interpretation that we will discuss in due time.

The results we are about to give concern both bosonic and fermionic systems, and the differences often lie only in sign changes between expressions. In fact, the rules behind Feynman diagrams are going to be the same, what will change however is the precise formula to evaluate such diagrams in a complete calculation.

2.4 Fermionic expectations

2.5 Perturbative series and Feynman diagrams

3 Explicit Calculation at Second Order

3.1 Duhamel's formula

Motivate the study of this system and the scalings $\hbar = N^{-1/3}$ and $\lambda = 1/N$.

The state space for a system of N identical fermions in a box of side length L with periodic boundary conditions is the space of completely antisymmetric L^2-functions on the torus $\mathbb{T}^{3N} = (\mathbb{R}^3/(2\pi\mathbb{Z}^3))^N$:

$$L^2_a(\mathbb{T}^{3N}) := \{\psi \in L^2(\mathbb{T}^{3N}) : \psi(x_{\sigma(1)}, ..., x_{\sigma(N)}) = \text{sgn}(\sigma)\psi(x_1, ..., x_N)\, \forall \sigma \in S_N\}$$

The Hamiltonian for weakly interacting particles can be written as a sum of the kinetic energy terms plus a sum of two-body interaction terms:

$$H = \hbar^2 \sum_{i=1}^{N} (-\Delta_{x_i}) + \lambda \sum_{1\leq i<j\leq N} V(x_i - x_j)$$

where $V : \mathbb{R}^3 \to \mathbb{R}$ is of class $\mathcal{C}^m$ for some $m \geq 1$ and with period L in all variables, and we assume moreover that $V(x_i - x_j) = V(|x_i - x_j|)$, which implies that $\hat{V}(-k) = \hat{V}(k)$. This fact will become useful in evaluating the second order perturbation to the ground state energy of the system. The scaling factor $\lambda \in \mathbb{R}$ is as discussed above. Working in the grand canonical picture requires to lift this operator to the fermionic Fock space, and requiring that the ground state energy is attained for a fixed number of particles that is greater than zero means that the Hamiltonian picks up an additional term $-\mu\mathcal{N}$, where μ is the constant called *chemical potential*. We absorb the chemical potential in the new definition of the free Hamiltonian as $H_0 = \sum_k (\hbar^2|k|^2 - \mu)a_k^* a_k$. Putting this all together, the Hamiltonian for this system is written as

$$H = H_0 + \lambda V = \sum_{k\in\mathbb{Z}^3} (\hbar^2|k|^2 - \mu)a_k^* a_k + \sum_{k,p,q\in\mathbb{Z}^3} \lambda\hat{V}(k) a^*_{p-k} a^*_{q+k} a_q a_p$$

where $\hat{V}(k)$ are the Fourier coefficients of V and $a_k^\#$ are creation/annihilation operators on the Fock space that create/annihilate states of definite momentum $k \in \mathbb{Z}^3$.
In the limit of zero temperature for a system of identical fermions, the chemical potential is also the value of the highest energy level occupied. We can therefore express it in terms of the number of particles in the ground state, as this parameter serves as a constraint for such a system through the relation (at zero temperature)

$$N = \sum_{k\in\mathbb{Z}^3} n_\text{F}(\epsilon_k) = \sum_{k\in\mathbb{Z}^3} \chi(\epsilon_k \leq \mu) = |\mathbb{B}_{k_\text{F}}(0) \cap \mathbb{Z}^3|$$

where k_F is the Fermi momentum, defined by $\hbar^2\epsilon_{k_\text{F}} = \mu$. We rederive the Fermi-Dirac distribution in Section (3.2), and with that expression we can see that, indeed, at zero temperature n_F is just a step function. This constraint is then just a counting problem for which we can use a well-known argument by Gauss, which involves giving as lower and upper bounds to the number of points on a lattice inside a sphere the number of unit cubes that are completely and at least partially inside the sphere, respectively. The result of such bounds can be summed up into the estimate given by the volume of the sphere plus an error proportional to its surface:

$$N = |\mathbb{B}_{k_\text{F}} \cap \mathbb{Z}^3| = \frac{4\pi}{3} k_\text{F}^3 + \mathcal{O}(k_\text{F}^2)$$

which implies that

$$k_{\mathrm{F}} = \left(\frac{3}{4\pi}N\right)^{\frac{1}{3}} + \mathcal{O}(N^0). \tag{10}$$

Since we want to compute the second order contribution to the ground state energy, we will calculate the free energy of the system and take its limit as the temperature goes to zero, that is $\beta \to \infty$. Recall that at thermal equilibrium the expectation value of any observable is given by the Gibbs state functional

$$\langle \cdot \rangle_\beta = \frac{\mathrm{Tr}(\cdot\, e^{-\beta H})}{\mathrm{Tr}(e^{-\beta H})}.$$

Because the free energy is given by the logarithm of the partition function divided by $-\beta$, we can make use of a useful formula that allows to truncate the quantity $\log(\mathrm{Tr}(e^{-\beta H}))$ to second order contributions in the scaling factor λ.

Proposition 3.1. *Given $H = H_0 + \lambda V$ as defined above, the following approximation for the free energy holds formally at second order λ, up to an overall factor of $-1/\beta$:*

$$\log \mathrm{Tr}\, e^{-\beta H} = \log \mathrm{Tr}\, e^{-\beta H_0} - \beta\lambda\langle V\rangle + \lambda^2 \int_0^\beta \mathrm{d}\tau \int_0^\tau \mathrm{d}\tau' \left(\langle V(\tau')V(\tau)\rangle - \langle V(\tau')\rangle\langle V(\tau)\rangle\right) \tag{11}$$

where Gibbs expectations are understood to be with respect to H_0 at temperature β.

Proof. Start by rewriting the operator $e^{-\beta H}e^{\beta H_0}$ using the fundamental theorem of calculus:

$$\begin{aligned}
e^{-\beta H}e^{\beta H_0} &= \mathbb{1} + \int_0^\beta \mathrm{d}\tau\Big(\frac{\mathrm{d}}{\mathrm{d}t}e^{-tH}e^{tH_0}\Big)(\tau) \\
&= \mathbb{1} + \int_0^\beta \mathrm{d}\tau e^{-\tau H}(-H + H_0)e^{\tau H_0} \\
&= \mathbb{1} + \int_0^\beta \mathrm{d}\tau e^{-\tau H}e^{\tau H_0}e^{-\tau H_0}(-\lambda V)e^{\tau H_0} \\
&= \mathbb{1} - \lambda \int_0^\beta \mathrm{d}\tau e^{-\tau H}e^{\tau H_0}V(\tau) = \star
\end{aligned}$$

Note that the integral is well defined as $\tau \mapsto \|e^{-\tau H}e^{\tau H_0}V(\tau)\|$ is a continuous function on the compact domain $[0,\beta]$ and is therefore Lebesgue integrable, which implies that $\tau \mapsto e^{-\tau H}e^{\tau H_0}V(\tau)$ is Bochner integrable. We briefly discuss integration of functions $\mathbb{R}^n \to \mathfrak{B}(\mathcal{H})$ in the appendix. Now we can see that in the integral we have the same term that appears in the l.h.s. but at a generic temperature τ. This means that we can repeat the same steps to push this operator to higher and higher “orders”. Iterating this procedure just once gives the following:

$$\begin{aligned}
\star &= \mathbb{1} - \lambda \int_0^\beta \mathrm{d}\tau \left(\mathbb{1} + \lambda \int_0^\tau \mathrm{d}\tau' e^{-\tau' H}e^{\tau' H_0}V(\tau')\right) V(\tau) \\
&= \mathbb{1} - \lambda \int_0^\beta \mathrm{d}\tau V(\tau) + \lambda^2 \int_0^\beta \mathrm{d}\tau \int_0^\tau \mathrm{d}\tau' V(\tau')V(\tau) + \mathrm{R}(\lambda, V, \beta).
\end{aligned}$$

where $\mathrm{R}(\lambda, V, \beta)$ is the remainder of this expression. Giving an estimate for the remainder that allows to prove the convergence of this iterative process is an entire problem on its own, so we will simply truncate the expression at second order and drop the remainder. Now we can invert this relation to isolate $e^{-\beta H}$ and take the trace. Since the trace is a continuous operator from trace class operators to $\mathbb{R}$ and $Ve^{-\beta H_0}$ is trace class, we can move it inside the integral (see Appendix A.1 and Appendix A.2) to get:

$$\begin{aligned}\operatorname{Tr} e^{-\beta H} &= \operatorname{Tr} e^{-\beta H_0} - \lambda \int_0^\beta \mathrm{d}\tau \operatorname{Tr}(V(\tau)e^{-\beta H_0}) + \lambda^2 \int_0^\beta \mathrm{d}\tau \int_0^\tau \mathrm{d}\tau' \operatorname{Tr}(V(\tau')V(\tau)e^{-\beta H_0}) \\ &= (\operatorname{Tr} e^{-\beta H_0}) \left(1 - \lambda \int_0^\beta \mathrm{d}\tau \, \langle V(\tau)\rangle + \lambda^2 \int_0^\beta \mathrm{d}\tau \int_0^\tau \mathrm{d}\tau' \, \langle (V(\tau')V(\tau)\rangle \right).\end{aligned}$$

We can now take the logarithm on both sides. Recall the Taylor expansion $\log(1+x) \sim x - x^2/2$ as $x \to 0$ and apply it to the r.h.s., but note that the first integral gives two contributions (one at first order and one at second order) while the second integral only contributes to the second order. In the end, we are left with:

$$\begin{aligned}\log \operatorname{Tr} e^{-\beta H} &= \log \operatorname{Tr} e^{-\beta H_0} - \lambda \int_0^\beta \mathrm{d}\tau \, \langle V(\tau)\rangle - \frac{1}{2}\left(\lambda \int_0^\beta \mathrm{d}\tau \langle V(\tau)\rangle\right)^2 \\ &\quad + \lambda^2 \int_0^\beta \mathrm{d}\tau \int_0^\tau \mathrm{d}\tau' \, \langle (V(\tau')V(\tau)\rangle \\ &= \log \operatorname{Tr} e^{-\beta H_0} - \beta\lambda\langle V\rangle \\ &\quad + \lambda^2 \int_0^\beta \mathrm{d}\tau \int_0^\tau \mathrm{d}\tau' \left(\langle V(\tau')V(\tau)\rangle - \langle V(\tau')\rangle\langle V(\tau)\rangle\right) \end{aligned} \tag{12}$$

where in the last step we performed the first integral and rewrote the square as a double integral, then taking into account the symmetry under $\tau \longleftrightarrow \tau'$ we could remove the factor $1/2$ by integrating only over the lower triangle $\{(\tau,\tau') \in [0,\beta]^2 : \tau' \leq \tau\}$. $\square$

3.2 Evaluating expectation values

To evaluate the expectation values, which are expectations of polynomials in field operators, we can turn to Grassmann variables. Indeed, by introducing a cutoff Λ we see that the possible momentum states form a finite set of M independent modes:

$$\mathcal{D}_\Lambda := \{k \in \mathbb{Z}^3 : |k| < \Lambda\} = \mathbb{Z}^3 \cap \mathbb{B}_\Lambda(0), \quad \text{with } |\mathcal{D}_\Lambda| = M.$$

Let then $\boldsymbol{\psi} = (\psi_1, ..., \psi_M)$ and $\bar{\boldsymbol{\psi}} = (\bar{\psi}_1, ..., \bar{\psi}_M)$ be the Grassmann variables associated to the families of operators a and a^* respectively. Having introduced the cutoff, the kinetic part of the Hamiltonian can now be written as a simple finite linear combination of fermionic number density operators, which makes it a bounded operator:

$$H_0 = \sum_{k \in \mathcal{D}_\Lambda} (\epsilon_k - \mu) a_k^* a_k$$

We begin the calculation by building the gaussian measure for the expectations $\langle \cdot \rangle_{H_0}$. Note that $\langle V \rangle_{H_0}$ contains field operators at the same temperature, while the expectations inside the double integral contain field operators at different temperatures, which means the two-point funcitons that appear in the contractions of the respective diagrams are going to be different.

Lemma 3.2. *The two-point function for this system has the following form:*

$$G_{k,k'}(\tau,\tau') = \langle a_k^*(\tau) a_{k'}(\tau')\rangle_{\beta,H_0} = \delta_{k,k'} n_{\mathrm{F}}(\epsilon_k) e^{-(\tau-\tau')(\epsilon_k-\mu)} \tag{13}$$

where $n_{\mathrm{F}}(\epsilon) = (e^{\beta(\epsilon-\mu)}+1)^{-1}$ *is the Fermi-Dirac distribution.*

Proof. We are going to divide the proof in two parts: first we will find an expression for the operators $a^{\#}(\tau)$, as this will allow us to write the two-point function in terms of $\langle a_k^* a_k' \rangle$, then we will compute this last expectation to find the Fermi-Dirac distribution. Recall that in the Heisenberg picture with imaginary time one has that for an operator A its evolution at temperature τ is given by $A(\tau) = e^{-\tau H_0} A e^{\tau H_0}$. Once again we will make use of the fundamental theorem of calculus to write

$$\begin{aligned} e^{-\tau H_0} a_k e^{\tau H_0} &= a_k + \int_0^\tau \mathrm{d}\lambda \, \frac{\mathrm{d}}{\mathrm{d}\lambda} \left(e^{-\lambda H_0} a_k e^{\lambda H_0} \right) \\ &= a_k + \int_0^\tau \mathrm{d}\lambda \, e^{-\lambda H_0} (-H_0 a_k + a_k H_0) e^{\lambda H_0} \\ &= a_k + \int_0^\tau \mathrm{d}\lambda \, e^{-\lambda H_0} [a_k, H_0] e^{\lambda H_0} \\ &= a_k + \int_0^\tau \mathrm{d}\lambda \, e^{-\lambda H_0} \sum_{k' \in \mathcal{D}_\Lambda} (\epsilon_{k'} - \mu) [a_k, a_{k'}^* a_{k'}] e^{\lambda H_0} = \star. \end{aligned}$$

Recall that, for operators A, B and C defined on a common domain, it holds that

$$[A, BC] = \{A, B\}C - B\{A, C\}$$

as can be shown by explicitly writing the expressions for the anticommutators. Using this identity we rewrite the commutator that appears in the last expression in terms of the canonical anticommutator as follows:

$$\begin{aligned} \star &= a_k + \int_0^\tau \mathrm{d}\lambda \, e^{-\lambda H_0} \sum_{k' \in \mathcal{D}_\Lambda} (\epsilon_{k'} - \mu)(\{a_k, a_{k'}^*\} a_{k'} - a_{k'}^* \{a_k, a_{k'}\}) e^{\lambda H_0} \\ &= a_k + \int_0^\tau \mathrm{d}\lambda \, e^{-\lambda H_0} (\epsilon_k - \mu) a_k e^{\lambda H_0} \\ &= a_k + (\epsilon_k - \mu) \int_0^\tau \mathrm{d}\lambda \, a_k(\lambda) \end{aligned}$$

where we used that $\{a_k, a_{k'}^*\} = \delta_{k,k'}$ and that $\{a_k, a_{k'}\} = 0$. We now have an expression for $a_k(\tau)$ in terms of its integral, and if we take the derivative with respect to τ we get the following Cauchy problem:

$$\begin{cases} \frac{\mathrm{d}}{\mathrm{d}\tau} a_k(\tau) = (\epsilon_k - \mu) a_k(\tau) \\ a_k(0) = a_k \end{cases} .$$

By the theorem of existence and uniqueness of the solution to an ODE on a closed interval, it suffices to find one explicitly. This is a well known differential equation and it's easy to check that the following is indeed a solution

$$a_k(\tau) = e^{\tau(\epsilon_k - \mu)} a_k(0) = e^{\tau(\epsilon_k - \mu)} a_k.$$

The procedure is completely analogous in the case of $a_k^*(\tau)$ and the result differs only by a sign in the exponent:

$$a_k^*(\tau) = e^{-\tau(\epsilon_k - \mu)} a_k^*(0) = e^{-\tau(\epsilon_k - \mu)} a_k^*.$$

By linearity, we can take the scalar exponential factors out of the expectation value to write the two-point function in terms of the quantity $\langle a_k^* a_{k'} \rangle$:

$$\langle a_k^*(\tau) a_{k'}(\tau') \rangle_{H_0} = e^{-\tau(\epsilon_k - \mu)} e^{\tau'(\epsilon_{k'} - \mu)} \langle a_k^* a_{k'} \rangle_{H_0}.$$

It remains to determine the expectation $\langle a_k^* a_{k'} \rangle$ and find the Fermi-Dirac distribution. Recall that the spectrum of the number density operators $a_k^* a_k$ is $\mathbb{N}_0$, and in the fermionic case the occupation number basis decomposes the Fock space like we discussed in Proposition 1.9. Using the cutoff on high momentum, the trace in the Gibbs state functional becomes a sum over a finite amount of states, therefore let $\{|\boldsymbol{n}\rangle = |n_1\rangle \otimes \cdots \otimes |n_M\rangle : n_i \in \{0,1\} \ \forall i = 1,\ldots,M\}$ be the complete orthonormal set of eigenvectors of M independent number density operators (in our case, $M = |\mathcal{D}_\Lambda|$). Calling $Z = \operatorname{Tr} e^{-\beta H_0}$ and using the momenta $k \in \mathcal{D}_\Lambda$ as indices for the basis states labels n_k, we write

$$
\begin{aligned}
\langle a_k^* a_{k'} \rangle_{\beta,H_0} &= \frac{\operatorname{Tr}(a_k^* a_{k'} e^{-\beta H_0})}{\operatorname{Tr} e^{-\beta H_0}} = \frac{1}{Z} \sum_{\boldsymbol{n}\in\{0,1\}^M} \langle \boldsymbol{n}| e^{-\beta H_0} a_k^* a_{k'} |\boldsymbol{n}\rangle \\
&= \delta_{k,k'} \frac{1}{Z} \sum_{\boldsymbol{n}\in\{0,1\}^M} \langle \boldsymbol{n}| a_k^* a_k |\boldsymbol{n}\rangle \prod_{p\in\mathcal{D}_\Lambda} e^{-\beta(\epsilon_p-\mu)n_p} \\
&= \delta_{k,k'} \frac{1}{Z} \sum_{\boldsymbol{n}\in\{0,1\}^M} n_k e^{-\beta(\epsilon_k-\mu)n_k} \prod_{p\neq k} e^{-\beta(\epsilon_p-\mu)n_p} \\
&= \delta_{k,k'} \frac{1}{Z} \left(\sum_{n_k=0}^{1} n_k e^{-\beta(\epsilon_k-\mu)n_k} \right) \prod_{p\neq k} \sum_{n_p=0}^{1} e^{-\beta(\epsilon_p-\mu)n_p} \\
&= \delta_{k,k'} \frac{1}{Z} e^{-\beta(\epsilon_k-\mu)} \prod_{p\neq k} (1 + e^{-\beta(\epsilon_p-\mu)})
\end{aligned}
$$

where we used the fact that H_0 is bounded (finite sum of bounded operators) and diagonal in the occupation number basis $\{|\boldsymbol{n}\rangle\}$, and that $a_k^* a_k |\boldsymbol{n}\rangle = n_k |\boldsymbol{n}\rangle$. Using many of the same steps for the denominator, we find that $Z = \prod_{p\in\mathcal{D}_\Lambda}(1 + e^{-\beta(\epsilon_p-\mu)})$, thus the expression simplyfies to

$$
\begin{aligned}
\langle a_k^* a_{k'} \rangle_{\beta,H_0} &= \delta_{k,k'} \frac{e^{-\beta(\epsilon_k-\mu)} \prod_{p\neq k}(1 + e^{-\beta(\epsilon_p-\mu)})}{\prod_p (1 + e^{-\beta(\epsilon_p-\mu)})} \\
&= \delta_{k,k'} \frac{e^{-\beta(\epsilon_k-\mu)}}{1 + e^{-\beta(\epsilon_k-\mu)}} \\
&= \delta_{k,k'} \frac{1}{e^{\beta(\epsilon_k-\mu)} + 1}.
\end{aligned}
$$

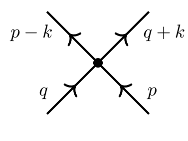
This is exactly the Fermi-Dirac distribution, and the thesis follows immediately. $\square$

With this quantity we can go over to the computation via Feynman diagrams. Recall the definition for the expectation of a function of Grassmann variables

$$
\mathbb{E}_G(\,\cdot\,) = \int P_G(\mathrm{d}\psi)(\,\cdot\,)\,, \quad \text{where } P_G(\mathrm{d}\psi) = \frac{1}{\det G} \left(\prod_{j=1}^{N} \mathrm{d}\psi_j \mathrm{d}\bar{\psi}_j \right) e^{-\bar{\boldsymbol{\psi}} G^{-1} \boldsymbol{\psi}}
$$

and the graphical representation of the interaction term via a vertex:

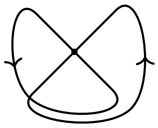
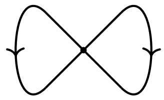
$$
V = \sum_{k,p,q} \hat{V}(k) \bar{\psi}_{p-k} \bar{\psi}_{q+k} \psi_q \psi_p =
$$



.

We can evaluate the expectations that show up in Duhamel's formula separately by following the diagrammatic rules we discussed in the previous chapter. Let's start with $\langle V \rangle_{H_0}$. This is the expectation of a sum of terms $\bar{\psi}\bar{\psi}\psi\psi$, exactly the interaction term, and so the result will be a sum of all possible (connected) diagrams with one vertex. Omitting the temperatures when writing two-point functions and calling $n_k = n_{\mathrm{F}}(\epsilon_k)$ the calculation is as follows, again omitting the subscripts β and H_0 for readability:

$$\langle V\rangle_{\beta,H_0} = \mathbb{E}_G\Bigg(\sum_{k,p,q} \hat{V}(k)\bar{\psi}_{p-k}\bar{\psi}_{q+k}\psi_q\psi_p\Bigg) = \sum_{\Gamma\in\mathcal{G}_{\mathrm{C},1}} \mathrm{val}(\Gamma)$$

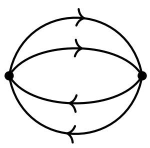
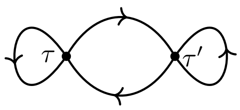
$=$  $+$ 

$$\begin{aligned} &= \sum_{k,p,q} \hat{V}(k)G_{p-k,p}G_{q+k,q} - \sum_{k,p,q} \hat{V}(k)G_{p-k,q}G_{q+k,p} \\ &= \sum_{k,p,q} \big(\hat{V}(k)\delta_{k,0}n_p n_q - \hat{V}(k)\delta_{k,p-q}n_p n_q\big) \\ &= \sum_{p,q} (\hat{V}(0) - \hat{V}(p-q))n_p n_q. \end{aligned}$$

Let us now consider the term $\langle V(\tau')V(\tau)\rangle - \langle V(\tau')\rangle\langle V(\tau)\rangle$ in the double integral now. This time we have terms with two vertices, and recalling the linked cluster theorem the expression factorizes as follows:

$$\begin{aligned} \langle V(\tau')V(\tau)\rangle - \langle V(\tau')\rangle\langle V(\tau)\rangle &= \mathbb{E}_G(V(\tau')V(\tau)) - \mathbb{E}_G(V(\tau'))\mathbb{E}_G(V(\tau)) \\ &= \mathbb{E}^{\mathrm{T}}_G(V(\tau'); V(\tau)) \\ &= \sum_{\Gamma\in\mathcal{G}_{\mathrm{C},2}} \mathrm{val}(\Gamma) = \star \end{aligned}$$

Assuming that V is such that $\hat{V}(-k) = \hat{V}(k)$, the connected diagrams with two vertices are exactly divided in two groups, up to a renaming of the indices in the resulting sums. Indeed, on one hand we have diagrams that have no loops, i.e. each vertex only has contractions with the other vertex and never with itself, and on the other we have diagrams with two loops. Consider the first group. Fixing an incoming and an outgoing leg on one vertex, there are 4 ways to contract them with legs of the other vertex, and in each case there is only one way to complete the diagram without introducing loops, therefore this group contains 4 diagrams. For the second group, there are 4 ways to form a loop on a vertex, and given a (connected) diagram with two vertices and two loops there exists only one way to complete it. Therefore the second group contains 16 diagrams (all possible ways to put one loop on both vertices). Accounting for these multiplicities, we have:

$\star = 4\ \ \tau$  $\tau'\ \ + 16$ 

$$\begin{aligned} &= \sum_{\substack{k,p,q\\k',p',q'}} \hat{V}(k')\hat{V}(k)\,(4\,G_{p-k,p'}G_{q'+k',q}G_{q+k,q'}G_{p'-k',p} + 16\,G_{p-k,q}G_{q+k,q'}G_{p'-k',p}G_{q'+k',p'}) \\ &= \sum_{p,q,p'} \Big(4\,(\hat{V}(p-p'))^2 e^{(\tau'-\tau)(\epsilon_p+\epsilon_q+\epsilon_{p'}+\epsilon_{p-p'+q}-4\mu)} n_p n_q n_{p'} n_{p-p'+q} \\ &\qquad + 16\,\hat{V}(0)\hat{V}(p-q)e^{(\tau'-\tau)(2\epsilon_p+\epsilon_q+\epsilon_{p'}-4\mu)} n_p^2 n_q n_{p'}\Big) \end{aligned}$$

Renaming some of the indices at both orders and integrating this last expression, as per equation (12), leads us to the following expression for the logarithm:

$$\begin{aligned}
\log \operatorname{Tr} e^{-\beta H} = \log \operatorname{Tr} e^{-\beta H_0} \\
&- \beta\lambda \sum_{p,k\in\mathcal{D}_\Lambda} \left(\hat{V}(0) - \hat{V}(k)\right) n_p n_{p-k} \\
&- \lambda^2 \sum_{p,q,k\in\mathcal{D}_\Lambda} \left(\frac{4(\hat{V}(k))^2 n_p n_q n_{q+k} n_{p-k}}{\epsilon_p + \epsilon_q + \epsilon_{q+k} + \epsilon_{p-k} - 4\mu} \left(\beta + \frac{1 - e^{-\beta(\epsilon_p+\epsilon_q+\epsilon_{q+k}+\epsilon_{p-k}-4\mu)}}{\epsilon_p + \epsilon_q + \epsilon_{q+k} + \epsilon_{p-k} - 4\mu} \right) \right. \\
&\left. + \frac{16\hat{V}(0)\hat{V}(p-q) n_p^2 n_q n_{p-k}}{2\epsilon_p + \epsilon_q + \epsilon_{p-k} - 4\mu} \left(\beta + \frac{1 - e^{-\beta(2\epsilon_p+\epsilon_q+\epsilon_{p-k}-4\mu)}}{2\epsilon_p + \epsilon_q + \epsilon_{p-k} - 4\mu} \right) \right). \qquad (14)
\end{aligned}$$

Now that the computation is complete, we need to prove that removing the UV cutoff, i.e. taking the limit $\Lambda \to \infty$, leads to a well-defined expression. Please note that this is not at all a trivial statement and what we are doing is deciding on the order in which to take different limits, as removing the cutoff starting from this expression means that the perturbative series is *not* done in the case where the state space is infinite-dimensional. Physically, this does not pose any concern and we expect the two scenarios to be equivalent, but from the point of view of mathematical rigour it is worth pointing out these kinds of subtleties.
So far we have been considering the kinetic energy to be $\epsilon_k = \hbar^2|k|^2$, but this expression is relevant specifically for the limit $\beta \to \infty$, while in the following theorem we will consider a more general form with one condition on the asymptotic behaviour at high momentum. The reason for this choice is that, at this level, we are not yet interested in the scaling of this expression in the number of particles, and having the possibility of proving the well-defined-ness of the perturbative series for systems that have a different kinetic term is worth the very little trouble this generalization costs. This is also partly motivated by the need to treat a special type of systems that can be treated via particle-hole transformations, a type of Bogolubov transformation, and we will comment on it later on.
A quick comment on a slightly less standard notation we will use to state some of the following results. We say that two real or complex sequences $\{x_n\}_n$ and $\{y_n\}_n$ are asymptotic if, and only if, $x_n/y_n \longrightarrow 1$, but from here we will mainly deal with quantities that are asymptotic only up to a proportionality constant. We will deal with the latter case by adopting the following notation:

$$x_n \asymp y_n \iff \exists C \neq 0 : \frac{x_n}{y_n} \longrightarrow C.$$

Lemma 3.3 (Non-stationary phase lemma). *If $m \in \mathbb{N}$ and $f : \mathbb{R}^n \to \mathbb{R}$ such that $\partial^\alpha f \in L^1(\mathbb{R}^M)$, for all multi-indices α of degree m, then for all such α*

$$|\hat{f}(k)| \leq \frac{1}{|k|^m} \int_{\mathbb{R}^n} \mathrm{d}x |\partial^\alpha f(x)|$$

Proof. ??? □

Theorem 3.4. *With the regularity condition that, given $m \geq 1$, $\partial^\alpha V \in L^1(\mathbb{T}^{3N})$ for all multi-indices α such that $|\alpha| = m$, and the assumption that ϵ_k depends only on $|k|$ and satisfies $\lim_{|k|\to\infty} \epsilon_k/|k| > 0$, the expression in Equation (14) converges in the limit $\Lambda \to \infty$.*

Proof. The zero-th order is trivial, as it doesn't depend on Λ and is, in fact, finite: $e^{-\beta H_0}$ is trace-class and strictly positive definite, which implies $|\log \operatorname{Tr} e^{-\beta H_0}| < \infty$. Moving onto the first and second orders, by the non-stationary phase argument we can conclude that $|\hat{V}(k)| \leq C_1/|k|^m$ for some $C_1 > 0$, which means that at both orders the factors $\hat{V}(k)$ will be decaying as an inverse power as $|k| \to \infty$, which is the relevant limit as we take the cutoff Λ to infinity. As for the energy terms, we are assuming that $\epsilon_k \geq C_2|k|$ for all k outside of the ball of radius K for some $K, C_2 > 0$. The last factor to discuss is the Fermi-Dirac distribution, but this has an explicit form: an exponential decay in ϵ_k, which is bounded at least by an explonential decay in $|k|$.

With these estimates in place we can show that, in the limit $\Lambda \to \infty$, the expression above converges. Looking at the first order, we can write

$$\begin{aligned}\sum_{p,k\in\mathcal{D}_\Lambda} \left|\left(\hat{V}(0) - \hat{V}(k)\right) n_p n_{p-k}\right| &\leq |\hat{V}(0)| \sum_{p,k\in\mathcal{D}_\Lambda} n_p n_{p-k} + \sum_{p,k\in\mathcal{D}_\Lambda} |\hat{V}(k)| n_p n_{p-k} \\ &= |\hat{V}(0)| \sum_{p\in\mathcal{D}_\Lambda} n_p \sum_{k\in\mathcal{D}_\Lambda} n_{p-k} + \sum_{p\in\mathcal{D}_\Lambda} n_p \sum_{k\in\mathcal{D}_\Lambda} |\hat{V}(k)| n_{p-k}.\end{aligned}$$

By the positivity of the general terms for all $k \in \mathbb{Z}^3$ we can bound the sums by the respective sum over $\mathbb{Z}^3$, and because the general terms of each of the above sums only depends on the modulus of the index we can employ the following strategy. Define $A_n = \{k \in \mathbb{Z}^3 : n \leq |k| < n+1\}$ and write

$$\sum_{k\in\mathbb{Z}^3} a_{|k|} = \sum_{n=0}^{\infty} \sum_{k\in A_n} a_{|k|}.$$

Because $c_1|A_n|a_{n+1} \leq \sum_{k\in A_n} a_{|k|} \leq c_2|A_{n+1}|a_n$ for all n if a_n is decreasing, and because $|A_n| \asymp n^2$, we can conclude that the behaviour of $\sum_{k\in\mathbb{Z}^3} a_{|k|}$ and $\sum_{n=0}^{\infty} n^2 a_n$ is the same. Now we can show that each term converges. Indeed $\sum_{k\in\mathbb{Z}^3} n_k$ converges if, and only if, the series $\sum_{\ell\in\mathbb{N}} \ell^2 e^{-\beta\epsilon_\ell}$ converges, by the asymptotic comparison test. The latter does converge absolutely as shown by the root test:

$$\left|\ell^2 e^{-\beta\epsilon_\ell}\right|^{1/\ell} = e^{\frac{2\log(\ell) - \beta\epsilon_\ell}{\ell}} \longrightarrow Q \in [0, 1).$$

Similarly, $\sum_{k\in\mathbb{Z}^3} |\hat{V}(k)| n_k$ converges, as

$$\sum_{k\in\mathbb{Z}^3} |\hat{V}(k)| n_k \leq \sum_{k\in\mathbb{Z}^3} \frac{C_1 n_k}{|k|^m} \leq \sum_{k\in\mathbb{Z}^3} C_1 n_k.$$

This shows that indeed the first order is, in the limit in $\Lambda \to \infty$, an absolutely convergent series.

Very similar arguments can be used to show that the second order has an absolutely convergent series as a limit. Indeed, recalling the assumptions made about each factor present in the expression, we can bound the first term in the sum (the second order direct term) by

$$\sum_{p,q,k\in\mathcal{D}_\Lambda} (\ldots) \leq \sum_p \frac{4n_p}{\epsilon_p - 4\mu} \sum_q n_q \sum_k (\hat{V}(k))^2 n_{q+k} n_{p-k} \left(\beta + \frac{1}{\epsilon_p - 4\mu}\right),$$

which, turning once more sums over $\mathbb{Z}^3$ into normal series with an additional square in the general term, is absolutely convergent, since the only true difference from the previous case is the additional factor $1/\epsilon_p$, which is an inverse power and does not change the result of any of the comparison tests done before. The exact same argument can be applied to the other term (second order exchange term), and this concludes the proof. $\square$

We can then replace the sums over $\mathcal{D}_\Lambda$ in equation 14 with series over $\mathbb{Z}^3$. Inserting the factor of $-1/\beta$ to get the free energy leads us to the complete expression for the free energy:

$$\begin{aligned} F &=: F_0 + F_{1,\mathrm{D}} + F_{1,\mathrm{X}} + F_{2,\mathrm{D}} + F_{2,\mathrm{X}} \\ &= F_0 + \lambda \sum_{p,k\in\mathbb{Z}^3} \hat{V}(0) n_p n_{p-k} - \lambda \sum_{p,k\in\mathbb{Z}^3} \hat{V}(k) n_p n_{p-k} \\ &\quad + \lambda^2 \sum_{p,q,k\in\mathbb{Z}^3} \frac{4(\hat{V}(k))^2 n_p n_q n_{q+k} n_{p-k}}{\epsilon_p + \epsilon_q + \epsilon_{q+k} + \epsilon_{p-k} - 4\mu} \left(1 + \beta^{-1} \frac{1 - e^{-\beta(\epsilon_p + \epsilon_q + \epsilon_{q+k} + \epsilon_{p-k} - 4\mu)}}{\epsilon_p + \epsilon_q + \epsilon_{q+k} + \epsilon_{p-k} - 4\mu}\right) \\ &\quad + \lambda^2 \sum_{p,q,k\in\mathbb{Z}^3} \frac{16\hat{V}(0)\hat{V}(p-q) n_p^2 n_q n_{p-k}}{2\epsilon_p + \epsilon_q + \epsilon_{p-k} - 4\mu} \left(1 + \beta^{-1} \frac{1 - e^{-\beta(2\epsilon_p + \epsilon_q + \epsilon_{p-k} - 4\mu)}}{2\epsilon_p + \epsilon_q + \epsilon_{p-k} - 4\mu}\right) \end{aligned} \tag{15}$$

where each term has been named appropriately following the order in which they appear: F_0 is the free energy of the unperturbed system (no interactions), $F_{i,\mathrm{D}}$ is the direct term at order i and $F_{i,\mathrm{X}}$ is the exchange term at order i.

3.3 Mean field scaling

With the convergence of the free energy under control, the next step is to look at the limit $\beta \to \infty$ to retrieve the ground state energy and complete our perturbative proof of the RPA scaling in the number of particles N.

Lemma 3.5. *With the same hypotheses as in theorem (3.4), all the series in Equation (15) converge uniformly in β over $[\delta, \infty)$ for some $\delta > 0$, and the limit $\beta \to \infty$ can thus be taken inside the sums.*

Proof. The only troubling factor is the Fermi-Dirac distribution, as it has a discontinuity at $\epsilon_k = \mu$ in the limit $\beta \to \infty$. By splitting each sum into a finite part up to, say, $\epsilon_k = 2\mu$, and a series over the remaining values of k, we can avoid this discontinuity. Moreover, since we are considering a limit away from $\beta = 0$ we can restrict the domain on which to check for uniform convergence to $[\delta, \infty)$ for some $\delta > 0$, and we'll see this is needed for some of the terms in the expression. With these things in mind we can see that the series $\sum_k n_k$ converges uniformly, as

$$\sum_{k:\epsilon_k \geq 2\mu} n_k \leq \sum_k \sup_{\beta\in[\delta,\infty)} n_k \leq \sum_k e^{-\delta(\epsilon_k - \mu)}$$

and the upper bound converges absolutely by the same arguments used in the previous theorem. Modifying this series with a prefactor of $1/(\epsilon_k - \mu)$, $\hat{V}(k)$ or $\hat{V}^2(k)$ in the general term does not change the result, as these factors do not depend on β and still lead to an absolutely convergent series as an upper bound. We can see that, by rewriting sums of the form $\sum_{p,q,k}(...)$ as $\sum_q(... \sum_p(... \sum_k(...)))$, we only need to check the convergence of the innermost series in all the terms of Equation (15), because the result we just proved shows that the rest will converge uniformly as it is going to be of the form $\sum_{p,q} C n_p n_q$ up to some inverse energy factors, where C is a constant that comes from the specific bound used for the innermost series. We can see this strategy in action for term F_1:

$$\begin{aligned} \sum_{p,k} \left|\hat{V}(0) - \hat{V}(k)\right| n_p n_{p-k} &= \sum_p n_p \sum_k \left|\hat{V}(0) - \hat{V}(k)\right| n_{p-k} \\ &\leq \sum_p n_p \sum_k \left|\hat{V}(0) - \hat{V}(k)\right| \sup_{\beta\in[\delta,\infty)} n_{p-k} = \star \end{aligned}$$

but the series containing $\sup_\beta(...)$ converges to some $C > 0$ as we discussed above, and we can therefore write $\star = \sum_p C n_p$, which again converges. For the direct term at second order, $F_{2,\mathrm{D}}$, following the same strategy of isolating the innermost series, we can identify two terms that need our attention, up to trivial bounds:

$$A := \sum_k \frac{\hat{V}(k)^2 n_{q+k} n_{p-k}}{\epsilon_{q+k} + \epsilon_{p-k} - 2\mu} \quad \text{and} \quad B := \sum_k \left| \frac{\hat{V}(k)^2 n_{q+k} n_{p-k}(1 - e^{-\beta(\epsilon_p + \epsilon_q + \epsilon_{q+k} + \epsilon_{p-k} - 4\mu)})}{\beta(\epsilon_{q+k} + \epsilon_{p-k} - 2\mu)^2} \right|$$

and the exchange term $F_{2,\mathrm{X}}$ has the same behaviour, both in terms of dependence on β and in terms of pointwise convergence. The series A is very simple: the Fermi-Dirac distributions are bounded by their supremum which allows us to show convergence by writing

$$A \leq \sum_k \frac{4\hat{V}(k)^2 e^{-\delta(\epsilon_{q+k} + \epsilon_{p-k} - 2\mu)}}{\epsilon_{q+k} + \epsilon_{p-k} - 2\mu}.$$

Let's now discuss B. Distributing the product at the numerator, the first term is a product of Fermi-Dirac distributions divided by β, which means it is monotonically decreasing in β for all q, p, k that respect the condition $\epsilon - \mu > 0$. As for the explonential term it is easy to check that, for all k far enough from the origin, the resulting general term is monotonically decreasing in β. Therefore, once again, the supremum is found at $\beta = \delta$ and we are left with

$$B \leq \sum_k \frac{\hat{V}(k)^2 e^{-\delta(\epsilon_{q+k} + \epsilon_{p-k} - 2\mu)}}{\delta(\epsilon_{q+k} + \epsilon_{p-k})^2} + \sum_k \frac{\hat{V}(k)^2 e^{-\delta(\epsilon_p + \epsilon_q + 2\epsilon_{q+k} + 2\epsilon_{p-k} - 6\mu)}}{\delta(\epsilon_{q+k} + \epsilon_{p-k})^2},$$

which shows that the second order is uniformly convergent too, completing the proof. □

Now that we have proved that we can take the limit of zero temperature inside the series, we are left with the task of computing it. The zero temperature Fermi-Dirac distribution is just a step function

$$n_{\mathrm{F}}(\epsilon) \xrightarrow[\beta \to \infty]{} \chi(\epsilon \leq \mu).$$

At last, using this fact we arrive at the ground state energy scaling in N, which is just a couple of estimates away.

Theorem 3.6. *Up to second order in perturbation theory, the ground state energy of a system of weakly interacting fermions exhibits the following scalings in the number of particles, N:*

- $F_0 \asymp N$,
- $F_{1,\mathrm{D}} \asymp N$ *while* $F_{1,\mathrm{X}} = \mathcal{O}(N^0)$,
- $F_{2,\mathrm{D}} \asymp N^{-\frac{1}{3}}$ *and also* $F_{2,\mathrm{X}} \asymp N^{-\frac{1}{3}}$ *CHECK SPECIFIC POWER OF N for* $F_{2,\mathrm{X}}$.

Proof. Of course we proceed order by order, starting from the unperturbed free energy to make sure that everything is in order. Recall that $B_{\mathrm{F}} = \{p \in \mathbb{Z}^3 : |p| < k_{\mathrm{F}}\}$ and expand the expression for F_0 as

$$F_0 = -\frac{\log(Z)}{\beta} = -\sum_{p \in \mathbb{Z}^3} \frac{1}{\beta} \log(1 + e^{-\beta(\epsilon_p - \mu)}) = \sum_{p \in B_{\mathrm{F}}} (\epsilon_p - \mu) + \sum_{p \notin B_{\mathrm{F}}} \log(1 + e^{-\beta(\epsilon_p - \mu)}),$$

where in the limit the second term contributes to order $\mathcal{O}(N^0)$. From here we can change the index summed over in $\sum_{|p| \leq CN^{1/3}}$ to approximate the result with an integral which contributes to order $\mathcal{O}(N^0)$. Recall that $\hbar = N^{-1/3}$ and $k_{\mathrm{F}} = \mathcal{O}(N^{1/3})$, then take $p \mapsto p' = N^{-1/3} p$, and since in the Fermi ball one has $|p| \leq CN^{1/3}$ this leads to $|p'| \leq C$. This allows for the following known approximation by integration:

$$\begin{aligned}
\sum_{p\in B_{\mathrm{F}}}(\hbar^2 p^2-\mu) &= N^{-\frac{2}{3}} \sum_{p\in\mathbb{Z}^3:|k|\leq CN^{1/3}} p^2 - \mu\sum_{p\in B_{\mathrm{F}}} 1 \\
&= N^{-\frac{2}{3}} N\left(\frac{1}{N}\sum_{p'\in\mathbb{Z}^3:|p'|\leq C}(N^{\frac{1}{3}}p)^2\right) - N\mu \\
&= N\left(\int_0^C p^2(4\pi p^2)\mathrm{d}p + \mathcal{O}(N^{-\frac{1}{3}})\right) - N\mu \\
&= \tilde{C}N + \mathcal{O}(N^{\frac{2}{3}})
\end{aligned}$$

where we used that, from Equation (10), $\mu = \hbar^2 k_{\mathrm{F}}^2 = \mathcal{O}(N^0)$. This shows that indeed $F_0 \asymp N$.

Onto the first order, recall the limit of the Fermi-Dirac distribution and the scaling $\lambda = N^{-1}$ to see that

$$\begin{aligned}
\lambda\sum_{p,k\in\mathbb{Z}^3}(\hat{V}(0)-\hat{V}(k))n_p n_{p-k} &= \lambda\sum_{p,k\in\mathbb{Z}^3}(\hat{V}(0)-\hat{V}(k))n_p n_k \\
&\xrightarrow[\beta\to\infty]{} N^{-1}\left(\sum_{p\in B_{\mathrm{F}}} 1\right)\left(\sum_{k\in B_{\mathrm{F}}}(\hat{V}(0)-\hat{V}(k))\right) \\
&= N\hat{V}(0) - \sum_{k\in B_{\mathrm{F}}}\hat{V}(k) \\
&\approx N\hat{V}(0) - \sum_{k\in\mathbb{Z}^3}\hat{V}(k).
\end{aligned}$$

where in the last steps we assumed a stronger regularity condition. In particular, we may choose to assume that $\hat{V}$ has compact support, or that $\hat{V}(k) \asymp e^{-|k|}$, so that the bound can provide an estimate that gets better as N grows. With this expression we can conclude that the first order indeed contributes to order $\mathcal{O}(N)$ with the direct term and to order $\mathcal{O}(N^0)$ with the exchange term.

To tackle the second order we use the same strategies that worked for the zero-th and first orders. We have to pay attention to the fact that, because of the product of Fermi-Dirac distributions $n_{q+k}n_{p-k}$ in both the direct and exchange terms, the innermost sums $\sum_k$ will contain the characteristic function of some set that is *contained* in a Fermi ball in the limit $\beta\to\infty$ (the set is an intersection of shifted Fermi balls). With this in mind, we can give an upper bound for this type of expression by summing over the entire Fermi ball and approximate the resulting sums with an integral. For the direct terms this is done as follows:

$$\begin{aligned}
&\lambda^2\sum_{p,q,k\in\mathbb{Z}^3}\frac{4\hat{V}(k)^2 n_q n_p n_{q+k} n_{p-k}}{\epsilon_q+\epsilon_p+\epsilon_{q+k}+\epsilon_{p-k}-4\mu} \\
&\quad\xrightarrow[\beta\to\infty]{} N^{-2}\frac{N}{N}\sum_{p,q\in B_{\mathrm{F}}}\sum_{\substack{k\in B_{\mathrm{F}}\\ q+k,p-k\in B_{\mathrm{F}}}}\frac{4\hat{V}(k)^2}{N^{-\frac{2}{3}}(q^2+p^2+(q+k)^2+(p-k)^2-4\mu N^{\frac{2}{3}})} \\
&\quad= N^{-1}\left(N^{\frac{2}{3}}\int_0^C \mathrm{d}p\mathrm{d}q\mathrm{d}k\frac{4\hat{V}(N^{1/3}k)^2(4\pi)^3p^2q^2k^2}{p^2+q^2+(q+N^{1/3}k)^2+(p-N^{1/3}k)^2-4\mu N^{2/3}} + \mathcal{O}(N^{-\frac{1}{3}})\right) \\
&\quad= \tilde{C}N^{-\frac{1}{3}} + \mathcal{O}(N^{-\frac{4}{3}}).
\end{aligned}$$

Note that, in the last line, $\tilde{C}$ techincally depends on N since the integrand has terms of the form $\hat{V}(N^{1/3}k)$, $N^{2/3}k^2$ and $\mu N^{2/3}$. This, however, does not pose any problem to us, because as $N\to\infty$

the integral will still converge (recall the additional regularity condition for V: either $\operatorname{supp}(\hat{V})$ is compact, or $\hat{V}(k) \sim e^{-|k|}$) and we may read $\tilde{C}N^{-1/3}+\mathcal{O}(N^{-4/3})$ as $(...) \asymp N^{-1/3}$ or even reabsorb an additional error $\mathcal{O}(N^{\cdots})$ from the integral into the other term $\mathcal{O}(N^{-4/3})$. The exchange term is really much of the same, with the only difference being that the integrand contains $\hat{V}(0)\hat{V}(N^{1/3}(p-q))$, which means the error that gets reabsorbed into $\mathcal{O}(N^{-4/3})$ will be slighlty different. □

4 On higher orders of perturbation theory

- comment on the regularity condition $\partial^\alpha \in L^1$ which excludes coulomb interaction;
- comment on third order somewhat explicitly;
- discuss the hope of proving that at every order n the direct term $\hat{V}(k)^n$ scales with $N^{-1/3}$;
- discuss difficulties in proving that the rest in duhamel's formula goes to zero and that non-direct terms converge.

A More about Hilbert spaces

We group in this appendix a series of definitions and theorems that are not necessarily part of the basic foundations of quantum mechanics, but are still crucial in understanding how exactly some computations are carried out and why they hold true. Some of the results presented here go beyond the scope of our work as far as their proof is concerned, and only serve the purpose of painting a clearer picture of what some tools actually are without just referencing an entire course on functional analysis.

A.1 Special operators

Definition A.1 (Self-adjoint operators)**.** An operator on a Hilbert space $A : D(A) \subseteq \mathcal{H} \to \mathcal{H}$ is called *self-adjoint* if $A = A^*$.

Note that this condition is stronger that being symmetric.

Definition A.2 (Compact operators)**.** An operator on a Hilbert space $A : D(A) \subseteq \mathcal{H} \to \mathcal{H}$ is called *compact* if $D(A) = \mathcal{H}$ and there exist complete orthonormal systems $\{u_i : i \in \mathbb{N}\}$ and $\{v_i : i \in \mathbb{N}\}$ for $\mathcal{H}$ and a sequence $\{\lambda_i : i \in \mathbb{N}\}$ such that $\lambda_i \to 0$ such that

$$A\phi = \sum_{n=0}^{\infty} \lambda_n \langle v_n, \phi \rangle u_n$$

for all $\phi \in \mathcal{H}$.

Definition A.3 (Trace class and Hilbert-Schmidt operators)**.** A compact operator A is said to be *trace class* if $\sum_{n=1}^{\infty} |\lambda_n| < \infty$ and is said to be *Hilbert-Schmidt* if $\sum_{n=1}^{\infty} |\lambda_n|^2 < \infty$. For a trace class operator A we define its *trace* as

$$\operatorname{Tr}(A) \coloneqq \sum_{n=1}^{\infty} \lambda_n \langle v_n, u_n \rangle.$$

Proposition A.1. *The trace of a trace class operator A is finite, and given an orthonormal basis $\{\phi_n\}_{n\in\mathbb{N}}$ for $\mathcal{H}$ the trace is*

$$\operatorname{Tr}(A) = \sum_{n=1}^{\infty} \langle \phi_n, A\phi_n \rangle.$$

Proof. ???? □

- $Ve^{-\beta H_0}$ is traceclass?
- comments on and PROPERTIES of $e^{-\beta H_0}$ with H_0 bounded and discussion on $Ve^{-\beta H_0}$ where V is bounded
- trace norm

Definition A.4 (Projection)**.** A linear operator P on a Hilbert space $\mathcal{H}$ which is self-adjoint ($P^* = P$) and idempotent ($P^2 = P$) is called a *projection operator* or *projector*.

Proposition A.2. *Given a Hilbert space $\mathcal{H}$, projectors defined on $\mathcal{H}$ have the following properties:*

a) every projector P is such that $P \neq 0$ and $\|P\| = 1$;

b) P is a projector if, and only if, $\mathbb{1} - P$ is a projector;

c) if P is a projector, then

$$\mathcal{H} = \mathrm{ran}P \oplus \ker P, \quad P|_{\mathrm{ran}P} = \mathbb{1}, \quad P|_{\ker P} = 0.$$

In particular, the spectrum of P is $\sigma(P) = \{0, 1\}$;

d) there is a bijection between projectors defined on $\mathcal{H}$ and closed subspaces of $\mathcal{H}$.

e) if $\{e_1, \dots, e_N\}$ for $N \in \mathbb{N}$ or $N = \infty$ is an orthonormal system in $\mathcal{H}$, then the projection P_M on $M = \mathrm{span}\{e_1, \dots, e_N\}$ is

$$P_M x = \sum_{n=1}^{N} \langle e_n, x\rangle e_n, \quad \forall x \in \mathcal{H}.$$

Definition A.5 (Isometry and Unitaries)**.** Given two Hilbert spaces $\mathcal{H}$ and $\mathcal{K}$, a linear map $U : \mathcal{H} \to \mathcal{K}$ with the property that

$$\|Ux\|_{\mathcal{K}} = \|x\|_{\mathcal{H}} \quad \forall x \in \mathcal{H}$$

is called an *isometry* between $\mathcal{H}$ and $\mathcal{K}$. A surjective isometry is called a *unitary operator*.

Unitary operators are of particular interest in quantum mechanics, as the property of preserving the inner product means that these are the natural objects to describe operations that need to preserve probability amplitudes.

Proposition A.3. *For a bounded linear operator between Hilbert spaces $U : \mathcal{H} \to \mathcal{K}$, the following statements are equivalent:*

a) U is a unitary operator;

b) $U^*U = \mathbb{1}_{\mathcal{K}}$ and $UU^* = \mathbb{1}_{\mathcal{H}}$;

c) $U\mathcal{H} = \mathcal{K}$ and $\langle x, y\rangle_{\mathcal{H}} = \langle Ux, Uy\rangle_{\mathcal{K}}$, for all $x, y \in \mathcal{H}$.

Recall that the dual of a complex Hilbert space $\mathcal{H}$ is the Hilbert space $\mathcal{H}^* = \{\varphi : \mathcal{H} \to \mathbb{C} : \varphi$ is a continuous linear functional$\}$, where continuity is with respect to the norm topology on $\mathcal{H}$. To conclude this part of our discussion on the basics of operators on a Hilbert space, we give three more results whose importance we will not discuss further, as it goes beyond the scope of our work.

Theorem A.4 (Riesz representation theorem)**.** *Let $\mathcal{H}$ be a complex Hilbert space. For every $x \in \mathcal{H}$ the map*

$$\begin{aligned} F : \mathcal{H} &\longrightarrow \mathbb{C} \\ y &\longmapsto F(y) := \langle x, y \rangle \end{aligned}$$

defines a continuous functional, and thus $F \in \mathcal{H}^*$. Conversely, for every $F \in \mathcal{H}^*$ there exists a unique $x_F \in \mathcal{H}$ such that $F(y) = \langle x_F, y \rangle$ for all $y \in \mathcal{H}$. Moreover, it holds that $\|x_F\|_{\mathcal{H}} = \|F\|_{\mathcal{H}^*}$.

Remark 22. The map $J : \mathcal{H} \to \mathcal{H}^*$ that sends a vector to its uniquely identifed continuous functional on $\mathcal{H}$ defines a canonical "isomorphism" between a Hilbert space and its dual, which means Hilbert spaces are self-dual. Technically J is not even a homomorphism since it is conjugate-linear rather than linear, and calling it an isomorphism is a bit imprecise, but this does not pose any problem to us.

Theorem A.5 (Spectral theorem for self-adjoint operators)**.** *Let $A : \mathcal{H} \to \mathcal{H}$ be a self-adjoint operator on the Hilbert space $\mathcal{H}$ over the field $\mathbb{K}$. Then there exist a measure space (X, Σ, μ), a unitary operator $U : L^2(X, \mathrm{d}\mu) \to \mathcal{H}$ and a function $f : X \to \mathbb{K}$ such that*

$$A = U M_f U^*$$

where M_f is the multiplication by f, i.e. the operator defined by $(M_f \psi)(x) = f(x)\psi(x)$.

Remark 23. In the case where A is also compact, then the spectrum $\sigma(A)$ is discrete and the function f could be seen as mapping an index over $\sigma(A)$ to an eigenvector, and the thesis would read as $(M_f \psi)_i = f_i \psi_i$, which tells us that ψ_i is an eigenvector with eigenvalue f_i. This special case makes it easier to understand this version of the spectral theorem as a generalization to the case of self-adjoint operators with continuous spectrum.

Theorem A.6 (C*-algebra of bounded operators)**.** *Let $\mathcal{H}$ be a Hilbert space. The set*

$$\mathfrak{B}(\mathcal{H}) = \{ A : \mathcal{H} \to \mathcal{H} : A \text{ is bounded } \}$$

is a Banach space, i.e. a complete normed space, via the usual definition of addition between operators (assuming the domain is the entire space $\mathcal{H}$) and multiplication by a scalar. It is also a C-algebra, i.e. a normed algebra with an involution $*$, via the adjoint of a bounded operator.*

A.2 Integration of operator-valued functions

We want to define integration for functions of the type $f : X \to \mathcal{B}$, where (X, Σ, μ) is a measure space and $\mathcal{B}$ is a Banach space (while differentiation is automatic if the measure space is $\mathbb{R}^n$ via the usual definition as limit of the incremental ratio). First we define *simple functions*: let $\{E_1, \dots, E_n\}$ be elements of the σ-algebra Σ and $\{b_1, \dots, b_n\}$ elements of $\mathcal{B}$, then the function

$$s(x) := \sum_{i=1}^{n} \chi_{E_i} b_i$$

is called a simple function and its integral is defined as

$$\int_X s \mathrm{d}\mu := \sum_{i=1}^{n} \mu(E_i) b_i.$$

Definition A.6 (Bochner integral)**.** A measurable function $f : X \to \mathcal{B}$ is called *Bochner integrable* if there exists a sequence of simple functions $\{s_n : n\mathbb{N}\}$ such that

$$\lim_{n \to \infty} \int_X \|f - s_n\|_{\mathcal{B}} \mathrm{d}\mu = 0,$$

i.e. if its point-wise norm is Lebesgue integrable. In this case, the *integral of f* is defined as

$$\int_X f \mathrm{d}\mu \coloneqq \lim_{n\to\infty} \int s_n \mathrm{d}\mu.$$

It can be shown that $\{\int_X s_n \mathrm{d}\mu : n \in \mathbb{N}\}$ is a Cauchy sequence, hence the integral is well-defined.

Remark 24. The integration rules for "nice" functions like powers and exponentials are identical to the case of scalar functions, just like for differentiation, but we will not prove this.

Proposition A.7. *Bochner integration satisfies the following properties:*

1. *a function $f : X \to \mathcal{B}$ is Bochner integrable if, and only if, $\int_X \|f\|_{\mathcal{B}} \mathrm{d}\mu < \infty$;*
2. *if $T : \mathcal{B} \to \mathcal{B}'$ is a continuous linear operator between Banach spaces and $f : X \to \mathcal{B}$ is Bochner integrable, then Tf is also Bochner integrable and it holds that*

$$T \int_E f \mathrm{d}\mu = \int_X T f \mathrm{d}\mu$$

   for all $E \in \Sigma$;
3. *(Dominated convergence theorem) if $\{f_n\}$ is a sequence of Bochner integrable functions that converges almost everywhere to a limit function f, and if there exists an integrable function g such that $\|f_n(x)\|_{\mathcal{B}} \leq g(x)$ almost everywhere on X, then it holds that*

$$\int_E \|f - f_n\|_{\mathcal{B}} \mathrm{d}\mu \to 0, \quad \textit{and} \quad \int_E f_n \mathrm{d}\mu \to \int_E f \mathrm{d}\mu$$

   for all $E \in \Sigma$;
4. *(Fundamental theorem of calculus) if $f : X \to \mathcal{B}$ is Bochner integrable over $[a,b] \subseteq \mathbb{R}$ and $F : X \to \mathcal{B}$ is an antiderivative for f, then*

$$\int_{[a,b]} f \mathrm{d}\mu = F(b) - F(a).$$

Remark 25. In particular, since functionals in the dual of a Hilbert space are also continuous operators between Banach spaces, given $f : X \to \mathcal{B}(\mathcal{H})$ a Bochner integrable function, it holds that

$$\langle \phi, \left(\int_X f \mathrm{d}\mu \right) \psi \rangle_{\mathcal{H}} = \int_X \langle \phi, f\psi \rangle_{\mathcal{H}} \mathrm{d}\mu.$$

Remark 26. An important property, which we make use of in our work, is that the set of trace-class operators is a Banach space with the trace norm, and the trace is a continuous operator from this space to the base field. This means that, because of the above proposition, whenever we take the trace of an integral of an operator-valued function, we can take the trace inside the integral.

With this notion of integration (and with the usual notion of differentiation) we can give an important theorem on ordinary differential equations that we used in our work. The proof of this theorem is completely analogous to the case where the Banach space is just $\mathbb{R}^n$, with the only difference being that the point-wise norm of a function is given by the norm on $\mathcal{B}$.

Theorem A.8 (Existence and uniqueness of a solution to an ODE)**.** *Let $f : [a,b] \times \mathcal{B} \to \mathcal{B}$ be a Lipschitz continuous function on $[a,b]$ uniformly on $\mathcal{B}$, and let it define the following Cauchy problem for $(x_0, y_0) \in [a,b] \times \mathcal{B}$:*

$$\begin{cases} y'(x) = f(x, y(x)) \\ y(x_0) = y_0 \end{cases}$$

where $y : [a,b] \to \mathcal{B}$. Then there exists a unique solution for the Cauchy problem over $[a,b]$.

A.3 Direct sums and tensor products of Hilbert spaces

Recall that, given two vector spaces V and W over a field $\mathbb{K}$, their *direct sum* is the set $V \times W$ endowed with a vector space structure defined via the following operations:

$$\begin{aligned}(v_1, w_1) + (v_2, w_2) &\coloneqq (v_1 + v_2, w_1 + w_2),\\ \lambda(v, w) &\coloneqq (\lambda v, \lambda w)\end{aligned}$$

for all $v, v_1, v_2 \in V$, $w, w_1, w_2 \in W$ and $\lambda \in \mathbb{K}$, and is denoted $V \oplus W$. Suppose now $\mathcal{H}_1$ and $\mathcal{H}_2$ are Hilbert spaces, then we have the natural result

Proposition A.9. *If $\langle \cdot, \cdot \rangle_i$ denotes the inner product on $\mathcal{H}_i$ for $i = 1, 2$, then the bilinear $\langle \cdot, \cdot \rangle$ given by*

$$\langle (x_1, x_2), (y_1, y_2) \rangle \coloneqq \langle x_1, y_1 \rangle_1 + \langle x_2, y_2 \rangle_2$$

defines an inner product with respect to which the space $\mathcal{H}_1 \oplus \mathcal{H}_2$ is complete. Thus $(\mathcal{H}_1 \oplus \mathcal{H}_2, \langle \cdot, \cdot \rangle)$ is a Hilbert space.

We can extend this definition of direct sum to the case where we consider a countable set of Hilber spaces $\{\mathcal{H}_i : i \in \mathbb{N}\}$ over the same field $\mathbb{K}$.

Proposition A.10. *The set $\mathcal{H}$ of sequences $x = \{x_i : x_i \in \mathcal{H}_i,\ i \in \mathbb{N}\}$ that satisfy*

$$\sum_{i=1}^{\infty} \|x_i\|_i^2 < \infty$$

is a Hilbert space with the operations $\{x_i\} + \{y_i\} \coloneqq \{x_i + y_i\}$, $\lambda\{x_i\} \coloneqq \{\lambda x_i\}$ and $\langle \{x_i\}, \{y_i\} \rangle \coloneqq \sum_{i=1}^{\infty} \langle x_i, y_i \rangle_i$. This Hilbert space is denoted

$$\mathcal{H} = \bigoplus_{i=1}^{\infty} \mathcal{H}_i.$$

To build the tensor product we will follow an algebraic approach which involves a similar procedure but with additional steps. Given two vector spaces V and W over a field $\mathbb{K}$, define the vector space of finite linear combinations of ordered pairs in $V \times W$ as

$$\Lambda(V, W) \coloneqq \left\{ \sum_{i=1}^{N} a_i (x_i, y_i) : a_i \in \mathbb{K},\ x_i \in V,\ y_i \in W\ \forall i = 1, \ldots, N \right\}.$$

On this space we impose algebraic relations through the quotien space

$$V \otimes W \coloneqq \Lambda(V, W) / \Lambda_0$$

where Λ_0 is the subspace generated by elements

$$\begin{aligned}&(x, y_1 + y_2) - (x, y_1) - (x, y_2),\\ &(x_1 + x_2, y) - (x_1, y) - (x_2, y),\\ &(\lambda x, y) - \lambda(x, y),\\ &(x, \lambda y) - \lambda(x, y)\end{aligned}$$

for all $x, x_1, x_2 \in V$ and $y, y_1, y_2 \in W$. If $\chi : \Lambda(V, W) \to \Lambda(V, W)/\Lambda_0$ is the projection to the quotient space, then we denote the elements in the quotient as $x \otimes y = \chi((x, y))$. Once again we can now extend this definition to the case of Hilbert spaces.

Proposition A.11. *Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be two Hilbert spaces over the same field. If $\langle \cdot, \cdot \rangle_i$ denotes the inner product on $\mathcal{H}_i$ for $i = 1, 2$, then the bilinear $\langle \cdot, \cdot \rangle$ given by*

$$\langle x_1 \otimes x_2, y_1 \otimes y_2 \rangle \coloneqq \langle x_1, y_1 \rangle_1 \langle x_2, y_2 \rangle_2$$

defines an inner product on $\mathcal{H}_1 \otimes \mathcal{H}_2$. The metric completion $(\mathcal{H}_1 \tilde{\otimes} \mathcal{H}_2, \langle \cdot, \cdot \rangle)$, unique up to isomorphism, is a Hilbert space, called tensor product of Hilbert spaces.

Remark 27. Note that, in general, the metric completion is required, as the algebraic tensor product by itself may not be complete with respect to the norm induced by the inner product defined in the proposition. However, with a slight abuse of notation we still denote $\mathcal{H}_1 \otimes \mathcal{H}_2$ the tensor product of the Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$.

In the case where $\mathcal{H}_1$ and $\mathcal{H}_2$ are separable Hilbert spaces, we can give an explicit construction of their tensor product. Indeed, let $\{u_i : i \in \mathbb{N}_1\}$ and $\{v_i : i \in \mathbb{N}\}$ be orthonormal systems for $\mathcal{H}_1$ and $\mathcal{H}_2$ respectively. Then it holds that

$$\mathcal{H}_1 \otimes \mathcal{H}_2 = \overline{\operatorname{span}\{u_i \otimes v_j : i, j \in \mathbb{N}\}},$$

which means that, in this scenario, the tensor product can be naturally identified with the closure of the space of finite linear combinations of elements $u \otimes v$, where $u \in \mathcal{H}_1$ and $v \in \mathcal{H}_2$, which are called *simple tensors* or *pure tensors*. Moreover, the set of simple tensors with factors taken from the respective complete orthonormal systems forms a complete orthonormal system for the tensor product.

Remark 28. With a simple inductive process one can conclude that the N-fold tensor product of Hilbert spaces is a Hilbert space, and the closure of the set of finite linear combinations of simple tensors still coincides with that tensor product.

Remark 29. A useful note for the type of computations we want to tackle in this work is that the space $L^2(\mathbb{R}^n) \otimes L^2(\mathbb{R}^m)$ can be naturally identified with the space $L^2(\mathbb{R}^{n+m})$, and given $f \in L^2(\mathbb{R}^n)$ and $g \in L^2(\mathbb{R}^m)$ one has that

$$f \otimes g(x, y) = f(x)g(y).$$

Lastly, we may define operators on a tensor product starting from operators on the factors. Indeed, let $A : D(A) \subseteq \mathcal{H}_1 \to \mathcal{H}_1$ and $B : D(B) \subseteq \mathcal{H}_2 \to \mathcal{H}_2$ be operators on the respective Hilbert space, then we can define their tensor product as

$$(A \otimes B)(\phi \otimes \psi) \coloneqq (A\phi) \otimes (B\psi)$$

on the set of pure pure tensors, i.e.

$$D(A \otimes B) = \{\phi \otimes \psi : \phi \in D(A),\ \psi \in D(B)\}.$$

[1]

References

[1] What. “wHAT”. In: *wHaT* (which year again?).